Characterization of coherent flow structures in brain ventricles

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Abstract

The dynamic flow of cerebrospinal fluid (CSF) in brain ventricles exhibits flow features on several scales, both spatially and temporally. Most analysis of this complex flow and the accompanying transport has used instantaneous (Eulerian) flow variables. Such analysis makes understanding of unsteady transport challenging. Here, we analyze brain ventricular CSF flow both in a Eulerian sense and from the Lagrangian perspective – a time-integrated view of the flow. With geometries generated from imaging data, we model CSF flow in adult human and embryonic zebrafish brain ventricles. In the human brain we model flow governed by cardiovascular pulsations, CSF secretion and motile cilia. The flow driven by cardiovascular pulsations is derived from a damped linear elastic model of brain ventricle deformations, as a result of applying displacement boundary conditions derived from experimental data. In the zebrafish brain we consider flow driven solely by motile cilia. The tissue and flow models are implemented and solved with finite element methods. We use the resulting velocity fields to compute finite-time Lyapunov exponent (FTLE) fields and use these fields to characterize Lagrangian coherent structures, which can be approximated by ridges in the FTLE fields. These coherent structures demonstrate prominent flow features in the brain ventricles congruent with findings in experimental research. In the human brain ventricles, we also investigate the role of inertia by comparing flow models governed by the Navier-Stokes and the Stokes equations. Comparisons show that solving the Stokes equations is adequate to compute integrated flow variables like stroke volumes, but that the Stokes approximation fails to resolve intricate features of flow and advective transport that are present in the solution to the Navier-Stokes equations, features that could be important to elucidating transport.

1 Introduction

Brains are constantly in motion. These motions are tightly coupled to the cardiac and respiratory cycles, bodily movement and neural activity. This brain motion is a key contributor to cerebrospinal fluid (CSF) flow (Balédent et al., 2001; Enzmann and Pelc, 1992; Sweetman and Linninger, 2011). CSF is produced by choroid plexus in the brain ventricles (Damkier et al., 2013), from where the fluid flows into the spinal cord or to the subarachnoid space (Rasmussen et al., 2018). The flow of CSF maintains brain homeostasis, delivers nutrients and removes waste, and serves as a cushion for the brain (Fame and Lehtinen, 2020; Kelley and Thomas, 2023; Linninger et al., 2016). Meanwhile, several pathologies afflicting the brain are related to disruption of CSF flow, such as hydrocephalus, where ventricular enlargement results from blockage of CSF flow (Linninger et al., 2016). The nature of CSF flow has been studied extensively in recent decades, both experimentally and computationally. Among the physiological processes that affect CSF flow are: blood vessel and respiration dynamics, secretion and absorption of CSF, beating of motile cilia, neural activity, and posture changes. (Balédent et al., 2001; Czosnyka et al., 2004; Faubel et al., 2016; Kelley and Thomas, 2023; Kurtcuoglu et al., 2007; Liu et al., 2024; Olstad et al., 2019; Sweetman and Linninger, 2011; Vinje et al., 2019). With these physical processes occurring over various time and length scales, CSF flow also consists of motions on different scales (Kelley and Thomas, 2023). Although CSF is fundamental to brain functioning, the underlying mechanisms of CSF flow are still not completely understood (Kelley and Thomas, 2023).

Because of the negative consequences of disruption in ventricular CSF flow, increasing our understanding of this flow is important to prevent or treat associated pathologies. In the brain ventricles, pulsations originating from cardiac and respiratory dynamics cause perturbations to blood volume and tissue deformation in the brain that result in CSF flow (Balédent et al., 2004; Enzmann and Pelc, 1991; Greitz et al., 1993, 1992; Kelley and Thomas, 2023; Liu et al., 2024; Vinje et al., 2019; Wagshul et al., 2011). Back-and-forth motion of CSF with vortical flow features has been visualized in human brains (Stadlbauer et al., 2010), and the pulsatility is clearly present in measurements of flow rates (Balédent et al., 2004; Liu et al., 2024; Vinje et al., 2019). At smaller scales, visualizations of ventricular CSF flow trajectories in animals have demonstrated intricate features such as stagnation and separation points, and vortex-structured flows created by motile cilia (Faubel et al., 2016; Gilpin et al., 2017; Olstad et al., 2019). Cilia-driven CSF flow has also to a lesser extent been studied in humans (Siyahhan et al., 2014; Worthington Jr and Cathcart III, 1963; Yoshida et al., 2022), demonstrating local effects of cilia on CSF flow patterns.

Flow patterns such as separatrices and vortices can be elucidated by special material structures known as Lagrangian coherent structures (LCS). The LCS constitute transport barriers in fluid advection, and organize stretching and folding in the mixing of passive tracers (Haller, 2015; Haller and Yuan, 2000). LCS can be characterized by computing finite-time Lyapunov exponent (FTLE) fields through integration of fluid parcel trajectories; ridges of these FTLE fields are good approximations of LCS (Shadden et al., 2005). Characterizing LCS thus provides a time-integrated view of fluid transport (Haller, 2015; Haller and Yuan, 2000; Shadden et al., 2005). The explanatory power of LCS has previously been demonstrated in a host of applications, including blood flow (Shadden et al., 2010; Shadden and Taylor, 2008), atmospheric flow (Günther et al., 2021), flow in airways (Lukens et al., 2010) and oceanic flows (Beron-Vera et al., 2015).

Despite the predictive power of studying flow from a Lagrangian perspective–where fluid parcel trajectories are tracked over time–only a limited amount of previous ventricular CSF flow studies have considered this perspective (Yoshida et al., 2022). Most studies are rather based on flow analysis with an Eulerian perspective (Causemann et al., 2022; Howden et al., 2008; Kurtcuoglu et al., 2007; Siyahhan et al., 2014; Sweetman et al., 2011), in which flow variables such as velocity and volumetric flux are measured at a fixed point in space and time. Instantaneous flow variables may however deceive interpretation of time-dependence in the flow (Shadden et al., 2005), especially in flows that contain motion at multiple scales, such as ventricular CSF flow. Therefore, further knowledge of flow and transport in brain ventricles, and the precise contributions of specific flow mechanisms such as cardiovascular pulsations, CSF secretion, and beating of motile cilia to mixing and transport of CSF, could be gained from Lagrangian analysis. For example, do LCS exist in brain ventricular CSF flow, and if so, how are the LCS organized? How are the LCS affected by flow mechanisms such as cardiovascular pulsations, motile cilia and CSF secretion? Does fluid inertia play a significant role in establishing the LCS and thus the advective transport within the brain ventricles?

In this work, we characterize LCS in brain ventricular CSF flow using two computational models generated with imaging data. One geometry represents the brain ventricles of an adult human. The other geometry represents the brain ventricles of embryonic zebrafish, an animal whose ventricular system shares many similarities with the human brain ventricles. Using finite element methods, we simulate CSF flow in these two geometries. The CSF velocity fields are used to compute particle trajectories. These trajectories are used to compute finite-time Lyapunov exponent (FTLE) fields, which we use to characterize LCS. In the human brain, we model flow driven by cardiovascular pulsations, choroid plexus secretion and motile cilia, and study the impact of each mechanism on flow patterns and advective transport. The results indicate the presence of LCS that are mostly determined by cardiovascular pulsations. For the zebrafish brain, we model flow driven by motile cilia, and demonstrate that FTLE fields can be used to characterize LCS that reflect flow trajectories previously observed in experiments. Altogether, our work demonstrates the usefulness of Lagrangian analysis in furthering our understanding of CSF flow.

2 Methods

2.1 Image-based computational meshes of ventricular geometries

To study brain ventricular CSF flow and characterize coherent flow structures, we consider two brain ventricular geometries generated with imaging data (Figure 1). One is the brain ventricular system of an adult human, which includes of the lateral, third and fourth ventricles (Figure 1A). The inter-ventricular foramina connect the lateral ventricles to the third ventricle, while the cerebral aqueduct connects the third and fourth ventricles (Figure 1A, B). In the lateral, third and fourth ventricles there are regions covered by choroid plexus tissue (Figure 1A). The lateral apertures and the median aperture serve as exit routes for the CSF. To represent the human ventricular system, we use a tetrahedral computational mesh extracted from the brain mesh originally generated by Causemann *et al.* (Causemann et al., 2022). This brain mesh was originally generated using surface geometries based on MRI images. Here, the standard version of the human brain ventricles mesh consists of 408 128 tetrahedral cells with maximal and minimal edge lengths of 0.40 cm and 0.023 cm, respectively.

The second geometry represents embryonic zebrafish brain ventricles at two days post-fertilization (2 dpf, Figure 1C). A surface geometry of these ventricles was in previous work generated with confocal microscopy (Olstad et al., 2019). This geometry was used to generate a tetrahedral computational mesh using fTetWild (Hu et al., 2020) (Figure 1D). At 2 dpf, the zebrafish brain ventricular system consists of three cavities: the anterior, middle and posterior ventricles (Figure 1C, D). The three compartments are connected by interventricular ducts. The computational mesh constitutes 132 134 tetrahedral cells, with a maximal (minimal) edge length of 14.9 μm (4.7 μm).

2.2 Governing equations of brain ventricular cerebrospinal fluid flow

We model cerebrospinal fluid (CSF) as an incompressible, Newtonian fluid of density $\rho = 1000$ kg/m^3 and dynamic viscosity $\mu = 0.7$ mPa $\cdot$ s (Bloomfield et al., 1998) in a domain $\mathcal{D} \subset \mathbb{R}^d$ with boundary $\partial\mathcal{D} \subset \mathbb{R}^{d-1}$. Neglecting the gravitational acceleration, no body forces are acting on the fluid. The flow is governed by the Navier-Stokes equations, which determine the fluid velocity $\boldsymbol{u}(\boldsymbol{x}, t)$ and pressure $p(\boldsymbol{x}, t)$ for times $t > 0$:

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = \nabla\cdot\mathbb{T} \quad \text{in } \mathcal{D}, \tag{1}$$

$$\nabla\cdot\boldsymbol{u} = 0 \qquad \text{in } \mathcal{D}, \tag{2}$$

with the stress tensor $\mathbb{T}$ defined as

$$\mathbb{T} = 2\mu\boldsymbol{\varepsilon}(\boldsymbol{u}) - p\mathbb{I}.$$

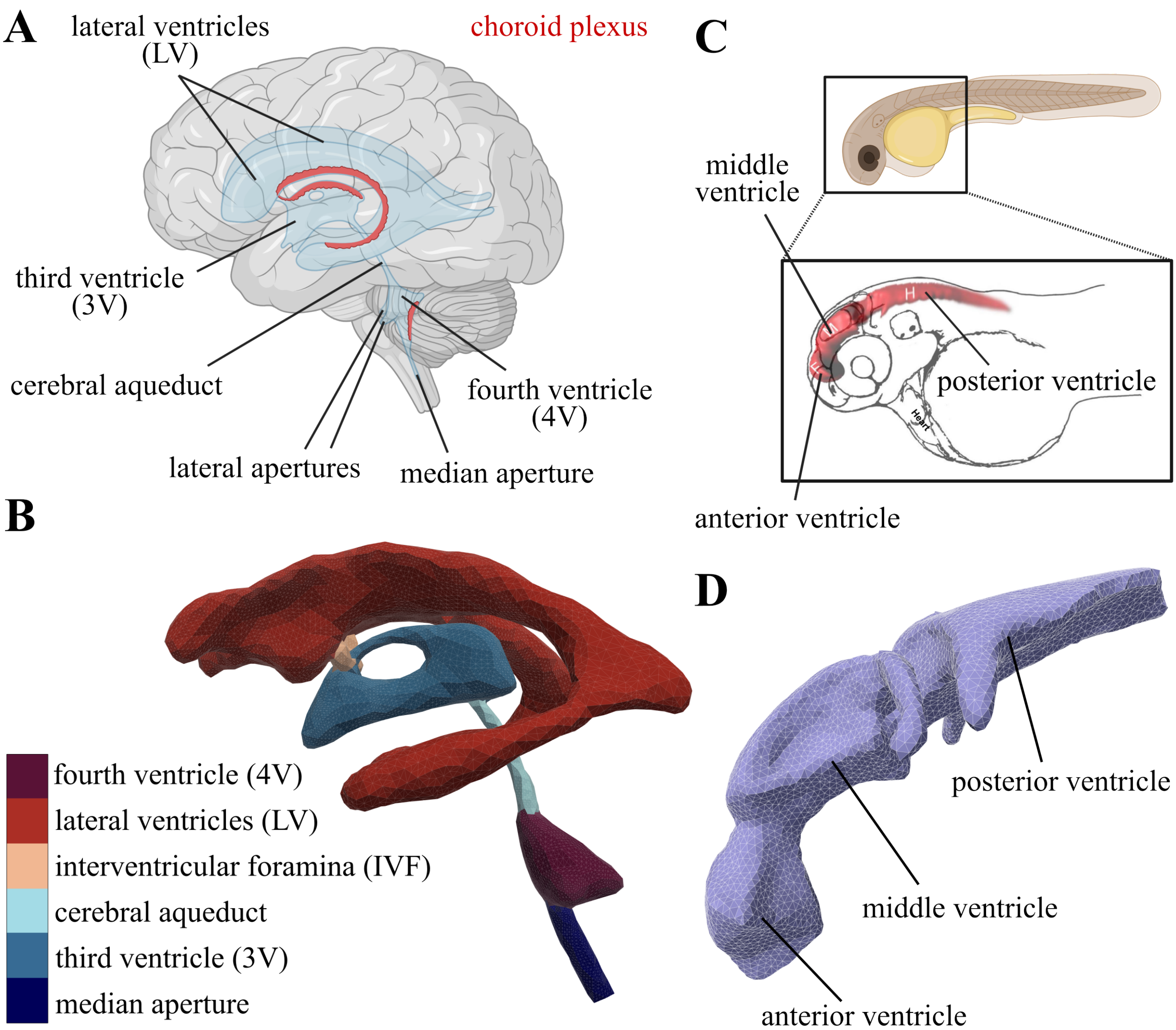

Figure 1: **A.** Illustration of the brain ventricular system in the adult human brain, with indication of the choroid plexus tissue (red structures) locations in the lateral, third and fourth ventricles. **B.** Computational mesh of human brain ventricles, sagittal view.. **C.** Illustration of the brain ventricular system in the embryonic zebrafish brain, indicating the location of the anterior, middle and posterior ventricles. **D.** Computational mesh of embryonic zebrafish brain ventricles, sagittal view. Parts of this figure were created with BioRender. The sketch of the zebrafish brain in panel C is adapted subject to the Creative Commons Attribution 4.0 International License (https://creativecommons.org/licenses/by/4.0/) (Korzh, 2018).

Here $\mathbb{I}$ is the identity tensor of dimension d and $\boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2}\left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T\right)$ is the strain-rate tensor.

For fluid flows where viscous forces dominate inertial forces, the steady Stokes equations, which read

$$-\nabla \cdot \mathbb{T} = \mathbf{0} \quad \text{in } \mathcal{D}, \tag{3}$$

are often used as an approximation to the momentum equation (Equation (1)). Viscous forces dominate in CSF flow in smaller brains such as embryonic zebrafish brain ventricles (Herlyng et al., 2025; Olstad et al., 2019), and therefore we use Equation (3) to model CSF flow in zebrafish. In the human brain, flow measurements and simulations indicate that CSF flow is in a laminar regime with non-negligible transient effects, but negligible inertial effects (Gholampour, 2018; Howden et al., 2008). In such a flow regime, the unsteady Stokes equations

$$\rho \frac{\partial \boldsymbol{u}}{\partial t} - \nabla \cdot \mathbb{T} = \mathbf{0} \quad \text{in } \mathcal{D}, \tag{4}$$

may serve as a fair approximation. In Equation (4), the convective acceleration term $\rho(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}$ in Equation (1) is neglected, while the local acceleration $\rho \partial \boldsymbol{u}/\partial t$ is retained to account for transient effects. Although most computational studies of CSF flow in human brain ventricles employ the Navier-Stokes momentum equation, some previous studies are based on Equation (4) (Causemann et al., 2022; Kelley and Thomas, 2023). We will compute flow in human brain ventricles with both the Navier-Stokes (Equation (1)) and the unsteady Stokes (Equation (4)) momentum equations to assess the role of fluid inertia, and whether the unsteady Stokes equations serve as a fair approximation in this setting.

2.3 Governing equations of brain tissue deformation

For the human brain ventricles, we use displacements of the brain ventricles walls as a driving force for CSF flow. The wall displacements are derived from a model of brain tissue deformation that considers the brain ventricles as a linear elastic material with viscoelastic behavior modeled by Rayleigh damping. The displacement $\boldsymbol{w}(\boldsymbol{x}, t)$ of the brain ventricles is then governed by the linear elastodynamics equations

$$\rho \ddot{\boldsymbol{w}} + \zeta_M \rho \dot{\boldsymbol{w}} - \zeta_K \nabla \cdot \mathbb{T}_t(\dot{\boldsymbol{w}}) - \nabla \cdot \mathbb{T}_t(\boldsymbol{w}) = \mathbf{0} \quad \text{in } \mathcal{D}, \tag{5}$$

where $\dot{\boldsymbol{w}} = \frac{\partial \boldsymbol{w}}{\partial t}$ and $\ddot{\boldsymbol{w}} = \frac{\partial^2 \boldsymbol{w}}{\partial t^2}$ are the displacement velocity and acceleration, respectively. The two damping factors ζ_M and ζ_K are related to inertia and stiffness. Moreover, $\mathbb{T}_t = 2\eta \boldsymbol{\varepsilon}(\boldsymbol{w}) + \lambda \operatorname{tr}(\boldsymbol{\varepsilon}(\boldsymbol{w}))\delta$ is the tissue stress tensor, in which η and λ are the Lamé parameters, and δ is the Kronecker delta.

The Lamé parameters η and λ are derived from Poisson's ratio ν and Young's modulus E of brain tissue. In Equation (5), the domain $\mathcal{D}$ corresponds to the fluid domain, so that a deformation field can be computed over this region and used to smoothly adjust the volumetric fluid mesh without large distortion of mesh elements near the boundary. Since this region is not tissue per se, we expect the precise choice of ν and E not to influence the general trends of the results, but rather only the required magnitude of the applied displacements $\boldsymbol{g}(\boldsymbol{x}, t)$. Additionally, there is in general an uncertainty in these material parameters. Especially E, for which values reported in the literature span several orders of magnitude (Budday et al., 2015; Cheng and Bilston, 2007, 2010; Sweetman et al., 2011). We adopt the choices of $\nu = 0.479$ and $E = 1500$ made by Causemann *et al.* (Causemann et al., 2022) in their computational model of pulsatile CSF flow in the brain. The ν value reflects a near incompressibility of brain tissue and the ventricular system, while the E value represents an average of gray and white matter brain tissue surrounding the ventricular system (Budday et al., 2015).

In general, the total Rayleigh damping factor ζ depends on two separate damping factors, one related to inertia (ζ_M) and one related to stiffness (ζ_K), and is given by $\zeta = \zeta_M/(2\omega) + \zeta_K \omega/2$ for some eigenfrequency ω (Chopra, 2007). The mass and stiffness damping factors $\eta_M = 0.20$ and $\eta_K = 0.10$ were set to dampen the motion such as to control vibrations around the cardiac cycle frequency 1 Hz, which is close to some of the eiqenfrequencies of the geometry.

2.4 Modeling CSF flow in deforming brain ventricles

The tissue displacements modeled by Equation (5) are used as a driving force for CSF flow by applying displacement velocities $\dot{\boldsymbol{w}}(\boldsymbol{x}, t)$ as boundary conditions in the fluid model. To account for the fact that the fluid then occupies a deforming domain, we solve the Navier-Stokes equations (Equations (1) and (2)) and the unsteady Stokes equations (Equations (2) and (4)) with an Arbitrary Lagrangian Eulerian (ALE) method (Duarte et al., 2004; Scovazzi and Hughes, 2007). The governing equations of CSF flow are solved in the reference configuration $\mathcal{D}_\chi = \mathcal{D}(\boldsymbol{x}, t_0)$, using an invertible mapping $\psi : \mathcal{D}_\chi \mapsto \mathcal{D}_{\boldsymbol{x}}$ between the reference configuration and the current (deformed) configuration $\mathcal{D}_{\boldsymbol{x}} = \mathcal{D}(\boldsymbol{x}, t)$. The mapping ψ induces a transformation of the differential operators in the governing fluid equations. Similarly, integrals of the weak formulation must be transformed appropriately to account for the domain motion (Scovazzi and Hughes, 2007).

2.5 Flow mechanisms and boundary conditions

CSF flow in adult human brain ventricles differs from CSF flow in embryonic zebrafish brain ventricles. Therefore, we consider different flow mechanisms and boundary conditions for the two applications. In the human brain, we consider three flow mechanisms:

- Deformation of the brain ventricles related to cardiovascular pulsatility, modeled by imposing a displacement velocity on the ventricular walls (see Section 2.5.1)
- Secretion of CSF from choroid plexus tissue, modeled by imposing a velocity normal to the wall which reflects a total influx of CSF (see Section 2.5.2)
- Ciliary beating, modeled by a tangential traction acting on the fluid at the boundary (see Section 2.5.3)

In the embryonic zebrafish brain ventricles, we consider ciliary beating as the only flow mechanism mainly for two reasons. First, an extensive description of cardiovascular pulsatility, similar to the one for human brain ventricles, does not exist (Herlyng et al., 2025). Second, in zebrafish choroid plexus secretion begins later than the developmental stage we consider (Bill et al., 2008).

The following three subsections detail our modeling approach for the three types of flow mechanisms outlined above. For a complete specification of the resulting CSF flow boundary conditions, see Section 2.5.4 (human model) and Section 2.5.5 (zebrafish model).

2.5.1 Modeling cardiovascular pulsation-related deformation of brain ventricles

The main driver of intraventricular CSF flow in humans is pulsatile volume displacement of the ventricles, which is closely related to the cardiac and respiratory cycles (Balédent et al., 2001; Neumaier et al., 2025; Sweetman and Linninger, 2011; Vinje et al., 2019; Wagshul et al., 2011). During systole, the blood vessels in the brain expand and concomitantly the parenchyma, putting pressure on the ventricular system. The inward displacement of the brain ventricles leads to antegrade (from cephalic to caudal) CSF flow. When the blood vessels relax and contract during diastole, the flow reverses and CSF flows back into the system (Greitz et al., 1992; Sweetman and Linninger, 2011). The respiratory cycle also influences CSF flow pressure gradients, but at different frequencies than the cardiac cycle (Vinje et al., 2019). In this study, the respiratory influence on brain deformations is not considered, and we model deformations of the ventricles with a cardiac cycle frequency.

We model CSF flow driven by cardiovascular pulsations through imposing a velocity boundary condition on the brain ventricular walls. This velocity boundary condition is given by the displacement velocities $\dot{\boldsymbol{w}}$ obtained by solving Equation (5). In our model, the displacement velocity determines the normal component of the CSF velocity

$$\boldsymbol{u} \cdot \boldsymbol{n} = \dot{\boldsymbol{w}} \cdot \boldsymbol{n} \quad \text{on } \mathcal{S}_1, \tag{6}$$

where $\boldsymbol{n}$ is the outward facing normal and $\mathcal{S}_1 \subset \mathcal{S} = \partial\mathcal{D}$.

Displacement boundary conditions based on experimental data are used to solve Equation (5) and determine $\dot{\boldsymbol{w}}$. To this end, we partition the boundary $\partial\mathcal{D}$ into four parts $\partial\mathcal{D} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \mathcal{B}_3 \cup \mathcal{B}_4$ (Figure 2A) and impose the boundary conditions

$$\boldsymbol{w} \cdot \boldsymbol{n} = \boldsymbol{g} \cdot \boldsymbol{n}, \quad \mathbb{T}_{t,\parallel} = \boldsymbol{0} \qquad \text{on } \mathcal{B}_1, \tag{7}$$

$$w_z = g_z, \qquad \mathbb{T}_t\boldsymbol{n} \cdot \boldsymbol{e}_x = \mathbb{T}_t\boldsymbol{n} \cdot \boldsymbol{e}_y = 0 \quad \text{on } \mathcal{B}_2, \tag{8}$$

$$\boldsymbol{w} = \boldsymbol{0} \qquad \text{on } \mathcal{B}_3, \tag{9}$$

$$\mathbb{T}_t\boldsymbol{n} = \boldsymbol{0} \qquad \text{on } \mathcal{B}_4, \tag{10}$$

where $\boldsymbol{g} := \boldsymbol{g}(\boldsymbol{x}, t) = (g_x, g_y, g_z)$ is determined from experimental data and $\boldsymbol{e}_i$ is the unit basis vector along the axis i. We have also here introduced the tangential traction vector $\mathbb{T}_\parallel = P_{\boldsymbol{n}}(\mathbb{T}\boldsymbol{n})$, where

$$P_{\boldsymbol{n}}(\boldsymbol{q}) = (\mathbb{I} - \boldsymbol{n} \otimes \boldsymbol{n})\boldsymbol{q},$$

is the projection of a vector $\boldsymbol{q}$ onto a plane normal to the boundary normal vector $\boldsymbol{n} := \boldsymbol{n}(\boldsymbol{x}, t)$. This projection thus yields a vector tangential to the boundary.

The boundaries $\mathcal{B}_1 \cup \mathcal{B}_2$ and imposed data consist of:

(i) The roof of the lateral ventricles, where corpus callosum displacement data is imposed;

(ii) The lateral sides of the frontal horns, where caudate nucleus displacement data is imposed;

(iii) The occipital and posterior horns, where we impose a systolic contraction followed by a diastolic expansion;

(iv) The walls of the third ventricle, where thalamus displacement data is imposed;

(v) The floors of the lateral and third ventricles, where we impose cephalocaudal motion.

For (i)–(iv) we impose motion normal to the boundary (Equation (7), green facets in Figure 2A), while for (v) only motion in the z direction—which aligns with the feet-head direction— is imposed (Equation (8), pink facets in Figure 2A).

The function $\boldsymbol{g}(\boldsymbol{x}, t)$ was determined by experimental data (Almudayni et al., 2023; Enzmann and Pelc, 1992; Greitz et al., 1992; Kurtcuoglu et al., 2007; Pahlavian et al., 2018; Soellinger et al., 2009, 2007), and we remark that the function is periodic in time with a period of one second (assumed to be one cardiac cycle). For some of the experimental data, directly applying the data produced deformations that resulted in physically unrealistic large amounts of CSF flow. Therefore, we reduced the magnitudes of some of the displacements, while retaining the qualitative behavior of a downward and contractile motion during systole and an upward and dilated motion during diastole. Namely, reduction of the magnitudes was calibrated by comparing the CSF flow results with experimental values of aqueductal stroke volume(Eide et al., 2021; Wagshul et al., 2011).

The boundary $\mathcal{B}_3$ consists of the median aperture outlet and the lateral apertures (yellow facets in Figure 2A). These boundaries are anchored for two reasons: First, to avoid rigid body motions when solving Equation (5); Second, because these boundaries are considered open boundaries in the fluid model, the displacement velocity $\dot{\boldsymbol{w}}$ on this boundary must be zero to avoid conflicting boundary conditions between the tissue and the fluid models. Finally, to let the remaining part of the boundary deform freely subject to the boundary conditions on $\mathcal{B}_1 \cup \mathcal{B}_2 \cup \mathcal{B}_3$, the boundary $\mathcal{B}_4$ is traction-free in all directions (blue facets in Figure 2A).

2.5.2 Modeling choroid plexus CSF secretion with a normal flux

In the human brain, CSF is mainly produced by secretion through the choroid plexus tissue in the lateral, third and fourth ventricles (Figure 1A, B) (Damkier et al., 2013). We model CSF secretion by imposing a volumetric flux normal to the choroid plexus boundary, which is denoted $\mathcal{S}_2 \subset \mathcal{S}$ (green facets in Figure 2B). We assume that all CSF that enters the brain ventricles is produced by the

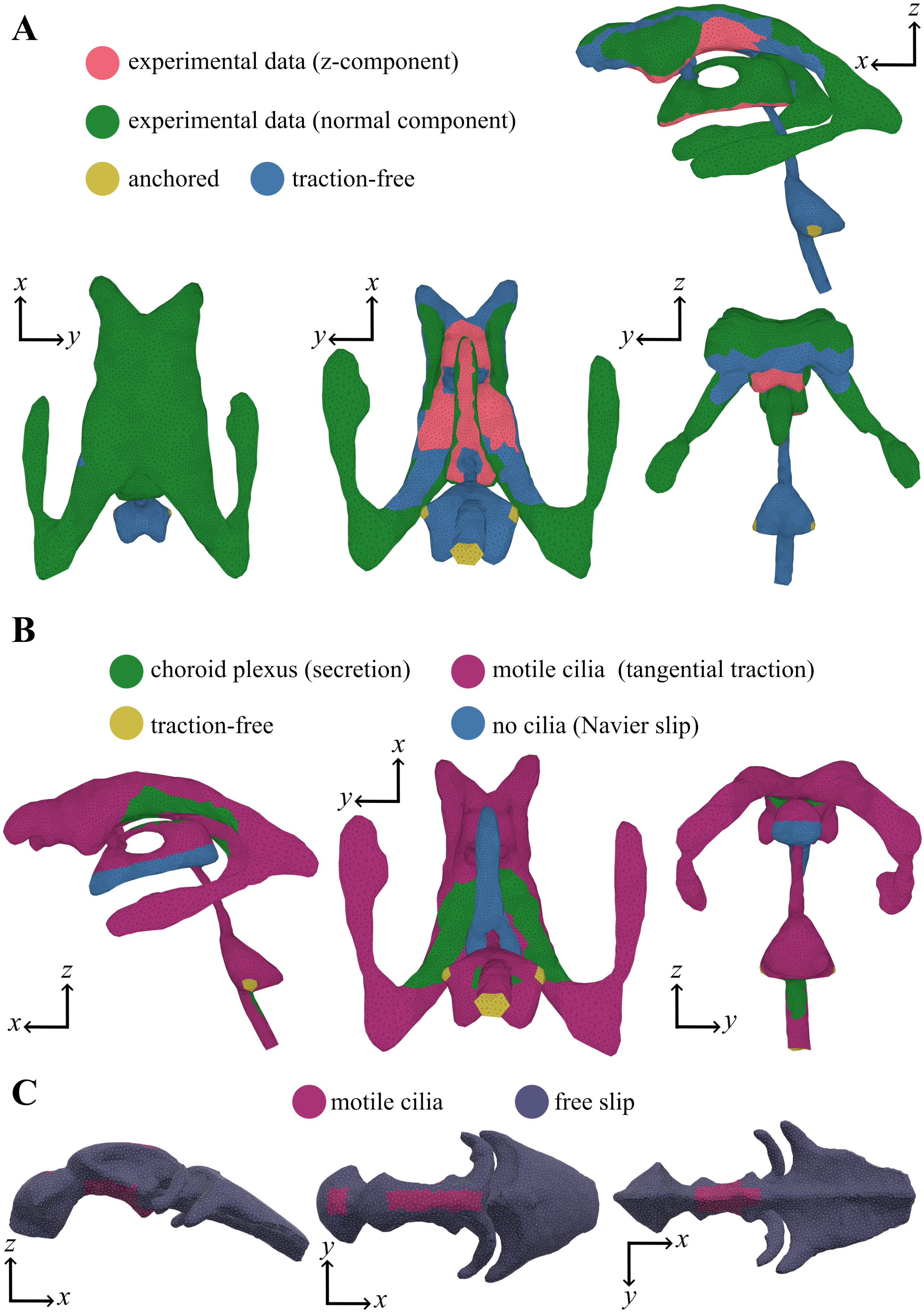

Figure 2: **A.** Human brain ventricles mesh with boundary facets colored by boundary condition type for the tissue deformation model. **B.** Human brain ventricles mesh with boundary facets colored by boundary condition type for the CSF flow model. **C.** Embryonic zebrafish brain ventricles mesh with boundary facets colored by boundary condition type for the CSF flow model.

choroid plexus, and that the influx is evenly distributed over the secretion sites in the lateral, third and fourth ventricles. A total daily CSF production of $Q_{\text{chp}} = 504$ ml/day is assumed (Czosnyka et al., 2004; Margetis et al., 2026). Since we assume even distribution at the different choroid plexuses, we impose

$$\boldsymbol{u} \cdot \boldsymbol{n} = -q_{\text{chp}} + \dot{\boldsymbol{w}} \cdot \boldsymbol{n} \quad \text{on } \mathcal{S}_2, \tag{11}$$

where $q_{\text{chp}} = Q_{\text{chp}}/S$ and S is the total choroid plexus boundary surface area. The normal displacement velocity $\dot{\boldsymbol{w}} \cdot \boldsymbol{n}$ is included to account for the deforming ventricular walls.

2.5.3 Modeling ciliary beating in the human brain by tangential traction

Motile cilia are hair-like structures that line the ependyma of brain ventricular walls, and the beating motion of cilia propel CSF (Olstad et al., 2019; Ringers et al., 2020). Motile cilia are abundant in the human brain (Coletti et al., 2018; Doetsch et al., 1997; Scott et al., 1973; Worthington Jr and Cathcart III, 1963), but are not found in the caudal third of the third ventricle (Bruni et al., 1972; Sánchez-Gomar et al., 2024; Scott et al., 1972) nor on choroid plexus (Ho et al., 2023; Narita and Takeda, 2015; Scott et al., 1972), although these latter areas may exhibit immotile cilia. Based on experimental research, we model cilia everywhere in the human brain ventricles, except for in the caudal third of the third ventricle and on choroid plexus (Bruni et al., 1972; Coletti et al., 2018; Doetsch et al., 1997; Ho et al., 2023; Narita and Takeda, 2015; Sánchez-Gomar et al., 2024; Scott et al., 1973, 1972,?; Worthington Jr and Cathcart III, 1963).

The cilia are modeled by imposing on the fluid a tangential traction $\boldsymbol{\tau}(\boldsymbol{x}, t) = \tau P_{\boldsymbol{n}}(\boldsymbol{r}(\boldsymbol{x}, t))$ that represents the collective beating motion of cilia, where τ is a constant that determines the magnitude of the tangential traction $\boldsymbol{\tau}$. In the human model, the tangential traction is imposed through a Navier slip boundary condition:

$$\mathbb{T}_{\parallel} = \boldsymbol{\tau}(\boldsymbol{x}, t) - \alpha P_{\boldsymbol{n}}(\boldsymbol{u}) \quad \text{on } \mathcal{S}_1 \setminus \mathcal{S}_4, \tag{12}$$

where the boundary $\mathcal{S}_4 \subset \mathcal{S}_1$ is the lower third of the third ventricle (blue facets in Figure 2B), where we do not impose cilia, and $\mathbb{T}_{\parallel} = P_{\boldsymbol{n}}(\mathbb{T}\boldsymbol{n})$ is the tangential traction vector. The friction coefficient $\alpha = \mu/L$ is determined by the dynamic viscosity μ and a slip length L. We model the cilia layer lining the ventricular walls as a slip layer, such that the fluid slips at the cilia while approaching a no-slip boundary condition at the underlying ventricular wall. We choose $L = 15\ \mu$m which is a typical length of a cilium in human brain ventricles (Afzelius, 2004; Siyahhan et al., 2014) as the slip length, resulting in $\alpha = 47$ Pa $\cdot$ s/m. In the regions where cilia are absent, we consider Equation (12) with $\boldsymbol{\tau}(\boldsymbol{x}, t) = \boldsymbol{0}$:

$$\mathbb{T}_{\parallel} = -\alpha P_{\boldsymbol{n}}(\boldsymbol{u}) \quad \text{on } \mathcal{S}_2 \cup \mathcal{S}_4. \tag{13}$$

Thus, although we assume motile cilia are absent on $\mathcal{S}_2 \cup \mathcal{S}_4$, we consider the ventricular walls of these regions to impose friction on the CSF. The regions $\mathcal{S}_2$ and $\mathcal{S}_4$ are illustrated by green and blue facets in Figure 2B, respectively.

For the human model, we model the direction $\boldsymbol{r}$ motivated by the work of Guirao *et al.*, who found the motile-cilia beating direction to be oriented by the CSF production during neural development (Guirao et al., 2010). To generate the vector $\boldsymbol{r}(\boldsymbol{x}, t)$, we simulate CSF flow in a production-only setup, with the choroid plexus boundary condition Equation (11), a zero normal traction condition on the median and lateral apertures, and $\boldsymbol{u} \cdot \boldsymbol{n} = 0$ on the rest of the boundary (that is, without wall displacements). The resulting velocity field, $\boldsymbol{u}_{\text{prod}}$, is normalized by its Euclidean length to yield the direction of the cilia:

$$\boldsymbol{r}(\boldsymbol{x}, t) = \frac{\boldsymbol{u}_{\text{prod}}(\boldsymbol{x}, t)}{|\boldsymbol{u}_{\text{prod}}(\boldsymbol{x}, t)|}.$$

There does not exist accurate data for the parameter τ that determines the magnitude of the tangential traction $\boldsymbol{\tau}$, with only a few computational studies reporting a value for a cilia force magnitude (Herlyng et al., 2025; Siyahhan et al., 2014; Thouvenin et al., 2020). None of these report values for humans. We set $\tau = 7.9$ mPa based on the force magnitude used by Siyahhan *et al.* (Siyahhan et al., 2014) for the purpose of comparison since they also modeled CSF flow in human brain ventricles, despite this value of τ being based on rodent brain data.

2.5.4 Summary of the boundary conditions for the governing CSF flow equations in human brain ventricles

For the human CSF flow model, the boundary is partitioned into four parts: $\partial\mathcal{D} = \mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2 \cup \mathcal{S}_3 \cup \mathcal{S}_4$. Using the mechanisms introduced in the previous subsections, the boundary conditions are

$$\boldsymbol{u} \cdot \boldsymbol{n} = \dot{\boldsymbol{w}} \cdot \boldsymbol{n} \quad \text{on } \mathcal{S}_1, \tag{14}$$

$$\mathbb{T}_{\|} = \tau P_{\boldsymbol{n}}(\boldsymbol{r}) - \alpha P_{\boldsymbol{n}}(\boldsymbol{u}) \quad \text{on } \mathcal{S}_1 \setminus \mathcal{S}_4, \tag{15}$$

$$\boldsymbol{u} \cdot \boldsymbol{n} = -q_{\text{chp}} + \dot{\boldsymbol{w}} \cdot \boldsymbol{n} \quad \text{on } \mathcal{S}_2, \tag{16}$$

$$\mathbb{T}_{\|} = -\alpha P_{\boldsymbol{n}}(\boldsymbol{u}) \quad \text{on } \mathcal{S}_2 \cup \mathcal{S}_4, \tag{17}$$

$$\mathbb{T}\boldsymbol{n} = \boldsymbol{0} \quad \text{on } \mathcal{S}_3. \tag{18}$$

We remark that the boundary $\mathcal{S}_4 \subset \mathcal{S}_1$ is the lower third of the third ventricle, where we do not impose the cilia tangential traction $\boldsymbol{\tau}$ as we consider motile cilia to be absent in these regions based on experimental observations. Moreover, $\dot{\boldsymbol{w}}(\boldsymbol{x}, t)$ is the displacement velocity of the ventricular walls, the vector $\mathbb{T}_{\|} = P_{\boldsymbol{n}}(\mathbb{T}\boldsymbol{n}) = \mathbb{T}\boldsymbol{n} - (\mathbb{T}\boldsymbol{n} \cdot \boldsymbol{n})\boldsymbol{n}$ is the tangential traction, and $P_{\boldsymbol{n}}(\cdot)$ is the projection of a vector onto a plane normal to the outward-facing normal vector $\boldsymbol{n}$. Equation (18) imposes zero traction, and is imposed on the end of the median aperture and on the lateral apertures to simulate open boundaries.

2.5.5 Boundary conditions for CSF flow in embryonic zebrafish brain ventricles

For the zebrafish model, the only flow mechanism we consider is motile cilia. We let the domain boundary be denoted $\mathcal{Z} = \mathcal{Z}_1 \cup \mathcal{Z}_2$. Similar to previous work (Herlyng et al., 2025), we consider a frictionless ($\alpha = 0$) tangential traction on the cilia boundary and a free-slip condition elsewhere:

$$\mathbb{T}_{\|} = \tilde{\boldsymbol{\tau}}(\boldsymbol{x}, t) \quad \text{on } \mathcal{Z}_1, \tag{19}$$

$$\mathbb{T}_{\|} = \boldsymbol{0} \quad \text{on } \mathcal{Z}_2, \tag{20}$$

$$\boldsymbol{u} \cdot \boldsymbol{n} = 0 \quad \text{on } \mathcal{Z}. \tag{21}$$

Condition (21) reflects the assumption that the ventricular walls are impermeable. The cilia boundary $\mathcal{Z}_1$ covers parts of the dorsal and ventral regions of the middle ventricle walls, as well as the dorsal wall of the anterior ventricle (Olstad et al., 2019) (purple facets in Figure 2C). The function $\tilde{\boldsymbol{\tau}}(\boldsymbol{x}, t)$ yields stronger cilia forces in the middle ventricle and weaker forces in the anterior ventricle, and was determined based on experimental data (Olstad et al., 2019). We refer the reader to Herlyng *et al.* (Herlyng et al., 2025) for more details.

2.6 Finite-time Lyapunov exponent fields and Lagrangian coherent structures

We computed finite-time Lyapunov exponent (FTLE) fields to identify Lagrangian coherent structures (LCS). LCS are transport barriers that partition regions of distinct flow behavior and act as organizing structures of fluid advection. LCS can be identified as ridges in the FTLE field (Shadden et al., 2005).

FTLE fields can be computed from fluid particle trajectory information (Shadden et al., 2005). Fluid particle trajectories $\boldsymbol{x}(t) \in \mathcal{D} \subset \mathbb{R}^d$ can be described by the dynamical system

$$\dot{\boldsymbol{x}}(t; t_0, \boldsymbol{x}_0) := \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{u}(\boldsymbol{x}(t; t_0, \boldsymbol{x}_0), t), \tag{22}$$

where t denotes time, $\boldsymbol{u}$ is the CSF velocity, and $\boldsymbol{x}_0 = \boldsymbol{x}(t_0; t_0, \boldsymbol{x}_0)$ is the initial position of a fluid particle at the initial time t_0. Equation (22) can be integrated to yield the particle position at a time t:

$$\boldsymbol{x}(t; t_0, \boldsymbol{x}_0) = \boldsymbol{x}(t_0) + \int_{t_0}^{t} \boldsymbol{u}(\boldsymbol{x}, \xi)\, \mathrm{d}\xi. \tag{23}$$

This relation can be used to define the flow map

$$\phi_{t_0}^t : \mathcal{D} \to \mathcal{D} : \boldsymbol{x}_0 \mapsto \phi_{t_0}^t(\boldsymbol{x}_0) = \boldsymbol{x}(t; t_0, \boldsymbol{x}_0). \tag{24}$$

Using the gradient of the flow map, we define the (right) Cauchy-Green deformation tensor

$$\mathbf{C} = \left(\nabla\phi_{t_0}^t\right)^* \nabla\phi_{t_0}^t, \tag{25}$$

where $\mathbf{A}^*$ denotes the transpose of a tensor $\mathbf{A}$. The finite-time Lyapunov exponent (FTLE) is defined as

$$\sigma := \sigma_{t_0}^{t_0+T}(\boldsymbol{x}_0) = \frac{1}{|T|} \ln \sqrt{\lambda_{\max}}, \tag{26}$$

where we have defined the integration time $T = t - t_0$, and $\lambda_{\max}$ is the maximum eigenvalue of $\mathbf{C}$ (Shadden et al., 2005). The FTLE measures how sensitive a fluid trajectory is to perturbations in its initial position. Namely, let $\boldsymbol{\delta}(t_0)$ be an infinitesimal change in position from a fluid particle at $\boldsymbol{x}_0$ at time t_0. This perturbation will have at most an exponential growth rate of σ according to (Shadden and Taylor, 2008)

$$\max ||\boldsymbol{\delta}(t_0 + T)|| \leq e^{\sigma|T|}||\boldsymbol{\delta}(t_0)||. \tag{27}$$

Note that the absolute value of the integration time T is employed because FTLE values can be computed both by integration forward ($t > t_0$) and backward ($t < t_0$) in time. FTLE values computed by integration forward in time reveal repelling LCS of the flow, whereas FTLE values computed by integration backward in time reveal attracting LCS.

2.7 Model scenarios and quantities of interest

To quantify flow and analyze the role of different flow mechanisms in the human brain model, we investigate three model scenarios: DSC includes all three flow mechanisms (deformation, secretion, cilia); DC includes deformation and cilia, excludes secretion; DS includes deformation and secretion, excludes cilia. To compare results for the different mechanisms, and also for comparison with previous computational and clinical studies, we compute the following quantities:

- The volumetric CSF flow rate

$$Q(t) = \int_{\mathcal{C}} \boldsymbol{u}(\boldsymbol{x}, t) \cdot \boldsymbol{n}(\boldsymbol{x}) \,\mathrm{d}S$$

  over different cross sections $\mathcal{C}$ in the interventricular foramina and cerebral aqueduct, as well as the outlets of the lateral apertures and median aperture.

- The cumulative CSF flow volume through the same cross sections $\mathcal{C}$ in the aqueduct, median aperture, lateral apertures and interventricular foramina. The cumulative flow volume $V(t)$ is computed by integrating the volumetric flow rate $Q(t)$ from an initial time t_0 to time t:

$$V(t) = \int_{t_0}^{t} Q(\xi) \,\mathrm{d}\xi.$$

- The CSF stroke volume in the aqueduct, median aperture, lateral apertures, and interventricular foramina. The stroke volume is the maximum (with respect to time) of the cumulative flow volume.

- The peak-to-nadir mean pressure gradient in the cerebral aqueduct, measured as the difference in the maximum and minimum value of the mean pressure gradient $\Delta\overline{p}$ over a cardiac cycle. Here, the mean denotes the cross-sectional average of the pressure:

$$\overline{p}(t) = \frac{1}{|\mathcal{C}|} \int_{\mathcal{C}} p(\boldsymbol{x}, t) \,\mathrm{d}S,$$

where $\mathcal{C}$ is a given cross section with area $|\mathcal{C}|$. Moreover, the Δ signifies the change in the mean pressure over the aqueduct length:

$$\Delta \overline{p} = \frac{\overline{p}_{\text{top}} - \overline{p}_{\text{bottom}}}{L_{\text{aq}}},$$

where "top" and "bottom" denote cross sections at the top and bottom of the aqueduct, while $L_{\text{aq}} = 23.5$ mm is the length that separates these cross sections.

- Maximum (u_{max}) and minimum (u_{min}) CSF velocity magnitudes over one cardiac cycle. At a given time t, the magnitude u of the velocity vector $\boldsymbol{u}(\boldsymbol{x}, t) = (u_x(\boldsymbol{x}, t), u_y(\boldsymbol{x}, t), u_z(\boldsymbol{x}, t))$ is

$$u(\boldsymbol{x}, t) = \sqrt{u_x^2 + u_y^2 + u_z^2}.$$

2.8 Numerical methods

This section provides details on the numerical methods used to solve the governing equations for brain tissue deformation and CSF flow, as well as the computation of FTLE fields. To summarize, the general algorithm of the computational model of the human brain ventricles is:

1. Solve the elasticity equations for the brain tissue displacements $\boldsymbol{w}(\boldsymbol{x}, t)$ and use $\boldsymbol{w}$ to compute the wall displacement velocity $\dot{\boldsymbol{w}}(\boldsymbol{x}, t)$, which is used as a boundary condition in the CSF flow model
2. Solve the fluid equations to obtain the CSF velocity field $\boldsymbol{u}(\boldsymbol{x}, t)$ and pressure $p(\boldsymbol{x}, t)$
3. Use the velocity $\boldsymbol{u}(\boldsymbol{x}, t)$ and pressure $p(\boldsymbol{x}, t)$ to compute quantities of interest
4. Use the velocity $\boldsymbol{u}(\boldsymbol{x}, t)$ to compute FTLE fields

The algorithm of the zebrafish model consists of steps 2 to 4.

2.8.1 Finite element method for the tissue deformation equations

The unsteady (damped) linear elasticity equations (Equation (5)) that govern the brain tissue deformation are discretized in space with a finite element method using fourth-order continuous Lagrange elements. For the displacement boundary conditions, we imposed traction-free conditions (Equation (10)) weakly. For the boundary conditions given by displacements $\boldsymbol{g}(\boldsymbol{x}, t)$ based on experimental data, the feet-head displacements (8) were imposed strongly, while the normal displacements (7) were imposed weakly using Nitsche's method (Benzaken et al., 2024; Nitsche, 1971). The anchor conditions (9) were imposed strongly.

In time, the tissue deformation equations are discretized with the Newmark-β method, for which updates of the displacement $\boldsymbol{w}$ and displacement velocity $\dot{\boldsymbol{w}}$ are

$$\begin{aligned} \boldsymbol{w}_{n+1} &= \boldsymbol{w}_n + \Delta t \dot{\boldsymbol{w}}_n + \left(\frac{1}{2} - \beta\right) \Delta t^2 \ddot{\boldsymbol{w}}_n + \beta \Delta t^2 \ddot{\boldsymbol{w}}_{n+1}, \\ \dot{\boldsymbol{w}}_{n+1} &= \dot{\boldsymbol{w}}_n + (1 - \gamma) \Delta t \ddot{\boldsymbol{w}}_n + \gamma \Delta t \ddot{\boldsymbol{w}}_{n+1}, \end{aligned} \tag{28}$$

where n denotes time step and $\Delta t = 0.001$ is the time step size, while $\ddot{\boldsymbol{w}}_n$ is the acceleration. We set $\gamma = 1/2$ and $\beta = 1/4$, which yields a second-order accurate and unconditionally stable scheme (Newmark, 1959). With this choice of γ and β, there is no numerical damping in the Newmark-β method. We observed high-frequency noise in the displacement velocity $\dot{\boldsymbol{w}}$ calculated with Equation (28), which is an issue that can arise when solving problems with prescribed displacement boundary conditions (Bathe and Noh, 2012). Therefore, to ensure smoothness of $\dot{\boldsymbol{w}}$ before applying it as a boundary condition in the fluid model, we computed $\dot{\boldsymbol{w}}$ by Fourier transformation of the displacement $\boldsymbol{w}$ and differentiation in the frequency space.

The applied displacement boundary conditions are periodic in time, with a period equal to one cardiac cycle (one second). To achieve periodicity in $\dot{\boldsymbol{w}}$, we do the following. The tissue deformation equations (5) are solved for four cardiac cycles, and the displacement velocities computed in the fourth cycle are used as boundary conditions in the fluid equations. Starting the system from rest with initial conditions $\boldsymbol{w}(\boldsymbol{x}, 0) = \dot{\boldsymbol{w}}(\boldsymbol{x}, 0) = \ddot{\boldsymbol{w}}(\boldsymbol{x}, 0) = \boldsymbol{0}$ and smoothly transitioning to the full periodic boundary data is important to achieve stable results, because an impulsive start could introduce non-physical oscillations. To ensure this smooth start, a cosine window is applied to the boundary condition magnitudes during the first half of the initial cardiac cycle. Additionally, we perform a ramp-up of the magnitudes over the first two cardiac cycles.

2.8.2 Finite element methods for the fluid equations

We discretize the governing equations of fluid motion, the Navier-Stokes (Equations (1) and (2)) and the Stokes (Equations (2) to (4)) equations, with $\boldsymbol{H}(\mathrm{div}; \mathcal{D})$-conforming finite element methods using Brezzi–Douglas–Marini (BDM) elements for the velocity and discontinuous Galerkin (DG) elements for the pressure (Brezzi et al., 1985; Nédélec, 1986). The BDM_k–DG_{k-1} elements, with polynomial order k, yield a spatial discretization that conserves mass exactly pointwise. For the unsteady Stokes equations, we use a weak formulation based on the one developed in Herlyng *et al.* (Herlyng et al., 2025). This weak form was further developed for the Navier-Stokes equations (Di Pietro and Ern, 2011; Schroeder et al., 2018).

For the BDM_k–DG_{k-1} elements, we use polynomial order $k = 2$ for the human model and $k = 1$ for the zebrafish model. The velocity component normal to facets is by definition continuous for the BDM elements, while tangential continuity is achieved by a stabilization technique (Hong et al., 2016). For the Navier-Stokes equations, the convective fluxes both on facets internal to the mesh and on open boundaries are stabilized with an upwinding technique (Di Pietro and Ern, 2011; Schroeder et al., 2018). The ventricular wall deformations are incorporated by formulating the governing fluid equations in an Arbitrary Lagrangian Eulerian framework (Duarte et al., 2004; Scovazzi and Hughes, 2007). In terms of boundary conditions, the cilia tangential traction boundary conditions and traction-free conditions are imposed weakly. The ventricular wall displacement velocity is imposed strongly. We use a Nitsche method to impose the choroid plexus flux boundary condition weakly (Benzaken et al., 2024; Nitsche, 1971).

In time, the fluid equations are discretized with the first-order, implicit Euler scheme using a time step $\Delta t = 0.001$ s. Since we use an implicit time discretization and solve the fluid equations directly, an initial condition is only needed for the velocity field, and not the pressure. When solving the Stokes equations, the fluid is initially at rest: $\boldsymbol{u}_0(\boldsymbol{x}) = \boldsymbol{u}(\boldsymbol{x}, 0) = \boldsymbol{0}$. For the Navier-Stokes equations, we solve the Stokes equations (with $\boldsymbol{u}_0(\boldsymbol{x}) = \boldsymbol{0}$) and use the solution $\boldsymbol{u}_{\mathrm{Stokes}}(\boldsymbol{x})$ as initial condition; $\boldsymbol{u}_0(\boldsymbol{x}) = \boldsymbol{u}_{\mathrm{Stokes}}(\boldsymbol{x})$. Moreover, when solving the Navier-Stokes equations the convective term is linearized as $(\boldsymbol{u}^{n+1} \cdot \nabla)\boldsymbol{u}^{n+1} \approx (\boldsymbol{u}^n \cdot \nabla)\boldsymbol{u}^{n+1}$, where n denotes time step. To achieve periodic flow in the unsteady simulations, the fluid equations are solved for two cardiac cycles and the velocities $\boldsymbol{u}(\boldsymbol{x}, t)$ of the second cycle are used to calculate FTLE fields. When reporting results, the start of this second cycle is defined as $t = 0$.

2.8.3 Finite-time Lyapunov exponent field computation

To compute finite-time Lyapunov exponent (FTLE) fields, we initiate a Cartesian grid of fluid particles at a release time t_0 and compute the flow map $\phi_{t_0}^{t_0+T}$ by solving Equation (23) with a fourth-order Runge-Kutta time stepping scheme. The time step size $\Delta t = 0.001$ s was chosen to satisfy the Courant-Friedrichs-Lewy condition $\Delta t \leq C h_{\min}/u_{\max}$, where $C = 0.5$, $h_{\min}$ is the minimal edge length of the mesh, and $u_{\max}$ is the maximum velocity magnitude. Further details on the FTLE computation algorithm is provided in previous work (Shadden et al., 2010). Because the FTLE computation algorithm interpolates velocity based on values in the mesh nodes, the finite element velocity approximations are projected from BDM_k into CG_k ($k = 2$ and $k = 1$ for the human and zebrafish model, respectively) and finally evaluated in a CG_1 space.

2.8.4 Software implementation

The FTLE field computations are performed with a framework that is implemented in C. The finite element methods for the fluid and tissue deformation equations are implemented and solved with DOLFINx (Baratta et al., 2023). A direct solver using MUMPS (Amestoy et al., 2011) is used to solve the fluid equations. We solve the tissue equations iteratively with a Flexible Generalized Minimal Residual (GMRES) method (Saad, 1993) preconditioned with the algebraic multigrid method BoomerAMG (Yang et al., 2002) from the hypre library (hypre, 2023). Both the FTLE computation software and the finite element method software are openly available and archived on Zenodo (Herlyng and Shadden, 2026).

3 Results

3.1 Pulsatile brain tissue deformation leads to pulsatile CSF flow in the human brain ventricles

When solving the tissue deformation equations with the applied experimental displacement data, during systole there is an immediate expansion of the third ventricle (3V) in concert with a cephalic movement of the 3V and the corpus callosum that lies superior to the lateral ventricles (Figure 3A, B). This is followed by a contraction of the whole ventricular system together with a caudal displacement of the brain ventricles. When solving the Navier-Stokes equations (Equation (1)) with the DSC model (deformations, secretion, cilia), the contraction and caudal displacement leads to a rapid increase in caudal CSF flow through the cerebral aqueduct and interventricular foramina (IVF, Figure 3C, D). In the aqueduct, the flow rate peaks at 0.22 ml/s after 26% of the cardiac cycle (Figure 3C). This is slightly earlier than the greatest volume displacement of the ventricular system in terms of magnitude; a volume change of -0.16% (contraction) that occurs 40% throughout the cardiac cycle (Figure 3A). The greatest positive volume displacement (expansion) occurs right after the onset of the cardiac cycle, constituting 0.01%. The lateral ventricles (LV) and the 3V are the main contributors to the volume change of the ventricular system. In comparison, the fourth ventricle (4V) volume changes are approximately a thousandth of the total volume change. The maximum displacement magnitude of 0.046 mm almost coincides with the largest volume displacement ($t = 0.41$ s), and was observed superior to the lateral ventricles on the right side ($\boldsymbol{x} = (15.7, -11.2, 29.9)$ mm).

When the ventricles relax during diastole (Figure 3A, B), the CSF flow reverses and flows cephalic to fill the ventricles and the aqueduct flow rate goes through a local minimum, increases slightly, then decreases again and bottoms out at -0.17 ml/s around 94% through the cycle (Figure 3C). These flow rate dynamics are similar in the IVF, and the total flow rate is evenly distributed through both of these foramina (Figure 3D). Finally, flow reverses when transitioning into the next cardiac cycle. Over the cardiac cycle, the flow rate magnitudes through the aqueduct, the lateral apertures, and both the IVF are all very similar (Figure 3D). These regions all have an order of magnitude greater flow rates than the flow rates through the median aperture. The maximum CSF velocity magnitude of caudally directed flow is 6.8 cm/s, which is located in the cerebral aqueduct during systole at 26% of the cardiac cycle. For cephalic flow, the minimum velocity is -4.4 cm/s in the aqueduct at the end of diastole after 94% of the cardiac cycle. The maximum velocity magnitude corresponds to a Reynolds number of 319, indicating laminar flow.

The gradient of the mean pressure (cross-sectional average) $\Delta\bar{p}$ along the cerebral aqueduct varies inversely with the flow rate behavior (Figure 3C), with a peak-to-through range of 0.053 mmHg/cm. The mean pressure at the superior end of the aqueduct oscillates over the cardiac cycle, going through three local minima and maxima (Figure 3E). The mean pressures in the IVF vary inversely with the aqueduct pressure, and the mean pressures in the two foramina are similar in magnitude. Over a cardiac cycle the stroke volume, which is the maximum of the cumulative flow volume, in the cerebral aqueduct was 46.1 μl (Figure 3F). Most of the displaced CSF volume crossing the domain boundary is through the lateral apertures with a stroke volume of 45 μl, with less fluid flowing in and out through the median aperture (stroke volume 8.6 μl). The flow volumes through the interventricular foramina connecting the lateral and third ventricles are evenly distributed between the two foramina (Figure 3D).

3.2 Flow fields predicted by the Stokes and Navier-Stokes equations are similar macroscopically, but differ locally

Solving the unsteady Stokes equations (Equation (4)) yields CSF flow that is qualitatively similar to the Navier-Stokes flow described in the previous section in terms of the systole and diastole trends. The Stokes approximation has the same aqueductal stroke volume (46.1 μl), but the displaced flow volumes through the lateral apertures (47 μl) and the median aperture (4.9 μl) differ from the Navier-Stokes solution, especially the median aperture stroke volume, which is 43% lower for the Stokes approximation. Although predicted maximum and minimum flow rates and velocities occur at virtually the same time instants, the Stokes equations predict lower maximum and minimum velocity magnitudes (6.0 and -4.6 cm/s, respectively). The 12% lower maximum velocity magnitude leads to a lower maximum Reynolds number of 282. The peak-to-nadir pressure gradient equals 0.054 mmHg/cm, slightly higher than that predicted by the Navier-Stokes equations.

Streamlines generated by integration forward in time at a given time instant with the Navier-Stokes velocity demonstrates that CSF flow is more directional at certain time instants, while having more sporadic directionality at other times (Figure 4A). Velocity vectors clearly show how the principle of mass conservation results in highest velocities through narrow sections like the cerebral aqueduct and interventricular foramina (Figure 4B). Velocities in larger compartments, like the lateral ventricles, are several orders of magnitude smaller. For example, during systole the velocities in the aqueduct reach 7.5 cm/s, while in the lateral ventricles the velocities are in the order of 0.1 cm/s. A close-up inside the third ventricle reveals disparate flow features for the Navier-Stokes and Stokes approximations (Figure 4C). The Navier-Stokes CSF velocity vectors have vortical patterns indicating a jet forming right above the entrance to the cerebral aqueduct (Figure 4C). These vortical patterns are absent in the Stokes solution.

3.3 Disregarding choroid plexus secretion or motile cilia lead to small changes in flow quantities

When comparing results for the three flow model versions (DSC, DC, DS), most of the computed flow quantities are similar. The aqueductal and lateral aperture stroke volumes are the same when excluding cilia (DS) as for the full model (DSC). Removing secretion while keeping cilia (DC model), there is slightly less flow with a 4% decrease in the aqueductal and lateral aperture stroke volumes when compared to DSC, with a similar decrease in the interventricular foramina stroke volumes. The median aperture stroke volume is less for both DS and DC compared to DSC, with 11% and 12% decreases for DS and DC, respectively. The pressure gradient peak-to-nadir is unaltered for both DC and DS. The maximal and minimal velocity magnitudes and flow rates change by less than 4% and 3% for DC and DS, respectively.

3.4 Motile cilia generate rotational flow in embryonic zebrafish brain ventricles

Solving the (steady-state) Stokes equations in the zebrafish brain ventricles, the motile cilia generate a rotational CSF flow with vortical structures confined to the different ventricular cavities (Figure 5, **a**). As a result of the applied tangential traction being constant and the geometry being closed, this flow field is a steady-state solution. The maximum velocity magnitude is 26.4 μm/s. Streamlines of particles integrated forward in time show rotational particle trajectories (Figure 5, **B**). In Figure 5, particles are seeded in the anterior, middle and posterior ventricles, and the three rotational structures originate from each of the particle seeding locations. Only seeding particles in e.g. the anterior ventricle would result in streamlines almost exclusively confined to the anterior ventricle, with only few particles crossing through the interventricular ducts.

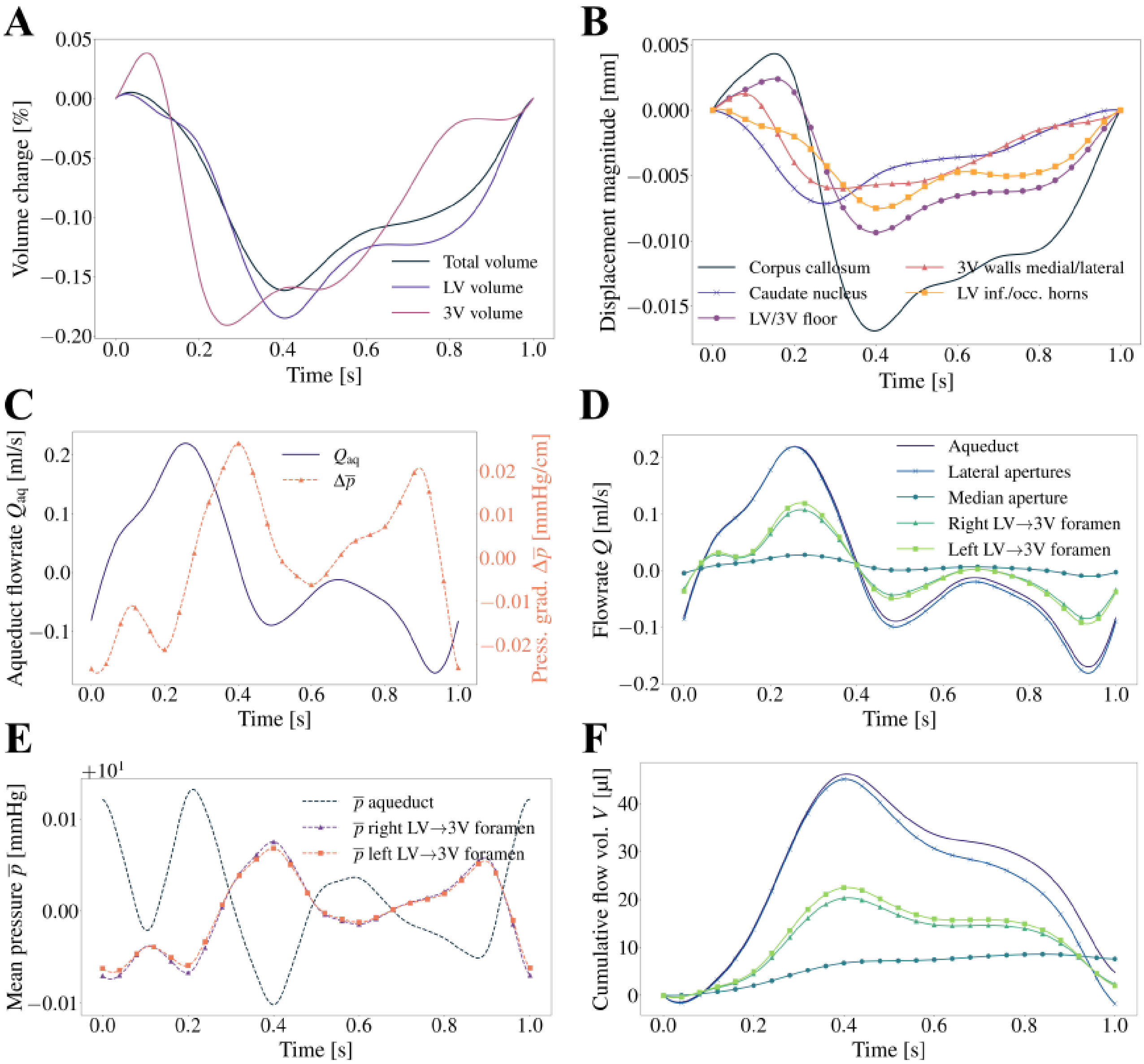

Figure 3: **A.** Volume dynamics of the whole ventricular system, the third ventricle (3V) and the lateral ventricles (LV) over a cardiac cycle. **B.** Illustration of the displacement boundary conditions $\boldsymbol{g}(\boldsymbol{x}, t)$ over a cardiac cycle (defined in Section 2.5.1). The maximum magnitudes of $\boldsymbol{g}(\boldsymbol{x}, t)$ are plotted for the five boundary condition regions: corpus callosum; caudate nucleus; lateral and third ventricle floor (LV/3V floor); third ventricle medial and lateral walls (3V walls medial/lateral); the inferior and occipital horns of the lateral ventricles (LV inf./occ. horns). **C–F.** Quantities of interest computed over one cardiac cycle: Flow rate $Q_{\mathrm{aq}}(t)$ and mean pressure gradient $\Delta\overline{p}(t)$ in the cerebral aqueduct (**C**); flow rates $Q(t)$ across cross sections in different parts of the ventricular system (**D**); mean pressures $\overline{p}(t)$ (**E**); cumulative flow volumes $V(t)$ (**F**). Common legend for panels **D** and **F**.

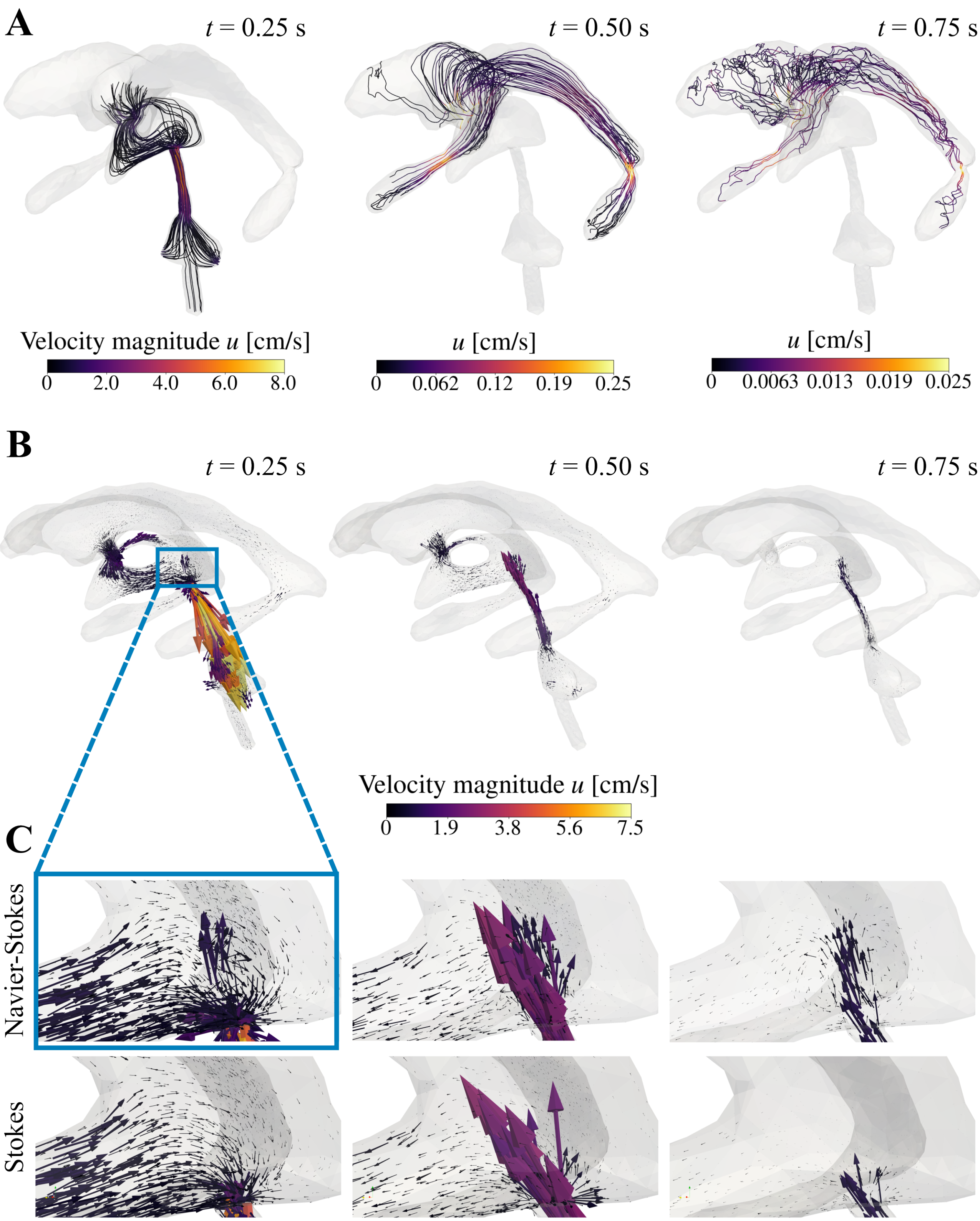

Figure 4: **A.** Forward-in-time integrated streamlines of particles seeded in the interventricular foramina at times $t = 0.25, 0.50, 0.75$ s, corresponding to systole, mid-cardiac cycle, and diastole, for the Navier-Stokes velocity approximation. Note the distinct color bar limit in each plot. **B.** Navier-Stokes velocity vectors at times $t = 0.25, 0.50, 0.75$ s in the human brain ventricles with zoomed-in view of (bottom row). **C.** Zoomed-in view (of the box region shown in the left-most subfigure of panel **B**) of velocity vectors superior to the cerebral aqueduct entrance, both for the Navier-Stokes and the Stokes solution, to highlight local differences in flow features.

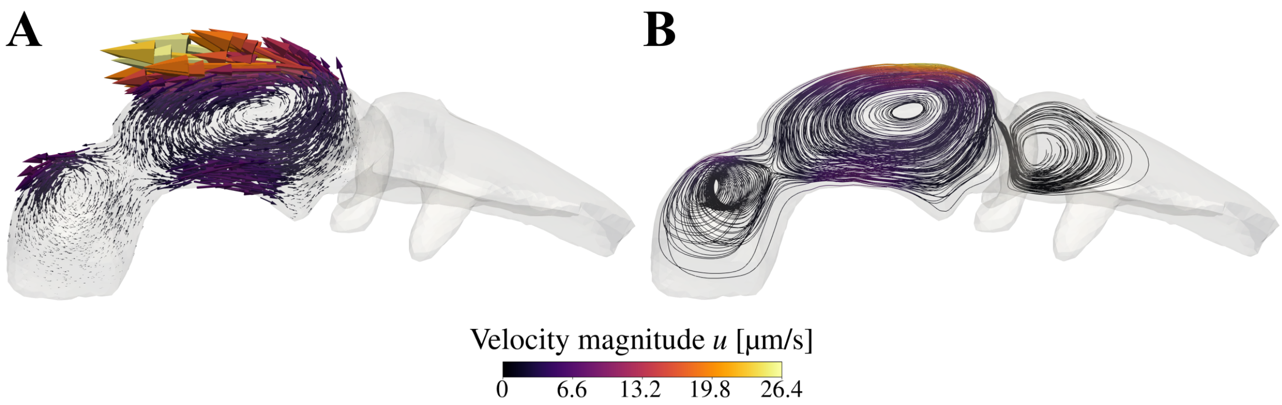

Figure 5: **A.** Velocity vectors scaled by magnitude. **B.** Streamlines of particles integrated forward in time, with particles seeded in the anterior, middle and posterior ventricles. Common color bar for **A** and **B**.

3.5 Finite-time Lyapunov exponent fields demonstrate prominent flow structures in the human brain ventricles

Distinct flow features (such as jets) have in previous computational studies been observed in the third ventricle at the cerebral aqueduct entrance (Kurtcuoglu et al., 2007). We therefore investigate structures in the Finite-time Lyapunov exponent (FTLE) fields near this entrance and in the interventricular foramina. The cerebral aqueduct and the interventricular foramina are among the narrowest sections of the ventricular system, where flow gradients increase as a result of the principle of mass conservation.

FTLE fields computed using the solution to the Navier-Stokes equations (Equation (1)) with varying integration time T show the emergence of a jet and the generation of a vortex ring at the cephalic entrance to the aqueduct (Figure 6A, B). The leading edge of the vortex ring is identified by ridges in the backward-in-time FTLE fields, ridges that develop with increasing T. In the forward-in-time FTLE field, there is a well-defined ridge for $T = 1.0$ s (one cardiac cycle) across the entrance to and the cerebral aqueduct (Figure 6A, B).

There are also clearly visible ridges that define lobe-like structures in the FTLE fields in the interventricular foramina (IVF) (Figure 6C). In the backward FTLE fields there develops a vortex ring that is ejected caudally. In the bulk of the lateral and third ventricles, there are no clear structures in the FTLE fields. We observe clearer structures as the integration time increases through a couple of cardiac cycles. For higher integration times, the structures become smeared out with diminishing FTLE values.

For an integration time of one cardiac cycle ($T = 1.0$ s), FTLE fields computed with varying initial release times t_0 show how the FTLE fields vary in time (Figure 7). In the IVF, the attracting structures reverse their orientation over the course of one cardiac cycle (one second, Figure 7A). In the 3V, the vortex formed by the jet near the aqueduct entrance detaches from the entrance during early systole (Figure 7B). A new vortex is forming during the transition to diastole, and a new vortex clearly forms during the transition to the next cardiac cycle.

3.6 When neglecting fluid inertia, the model fails to resolve important Lagrangian structures

When computing FTLE fields with the unsteady Stokes equations (Equation (4)), the jet structure in the third ventricle vanishes (Figure 8A). This differs greatly from the structure observed in the Navier-Stokes solution (Figure 8B). In the backward FTLE fields, the lobes that form and over time propagate away from the aqueduct entrance in the Navier-Stokes solution are not present in the Stokes solution's FTLE fields. This indicates a stronger stirring effect in the Navier-Stokes solution. Similarly, the repelling structures are more extensive in the Navier-Stokes FTLE fields.

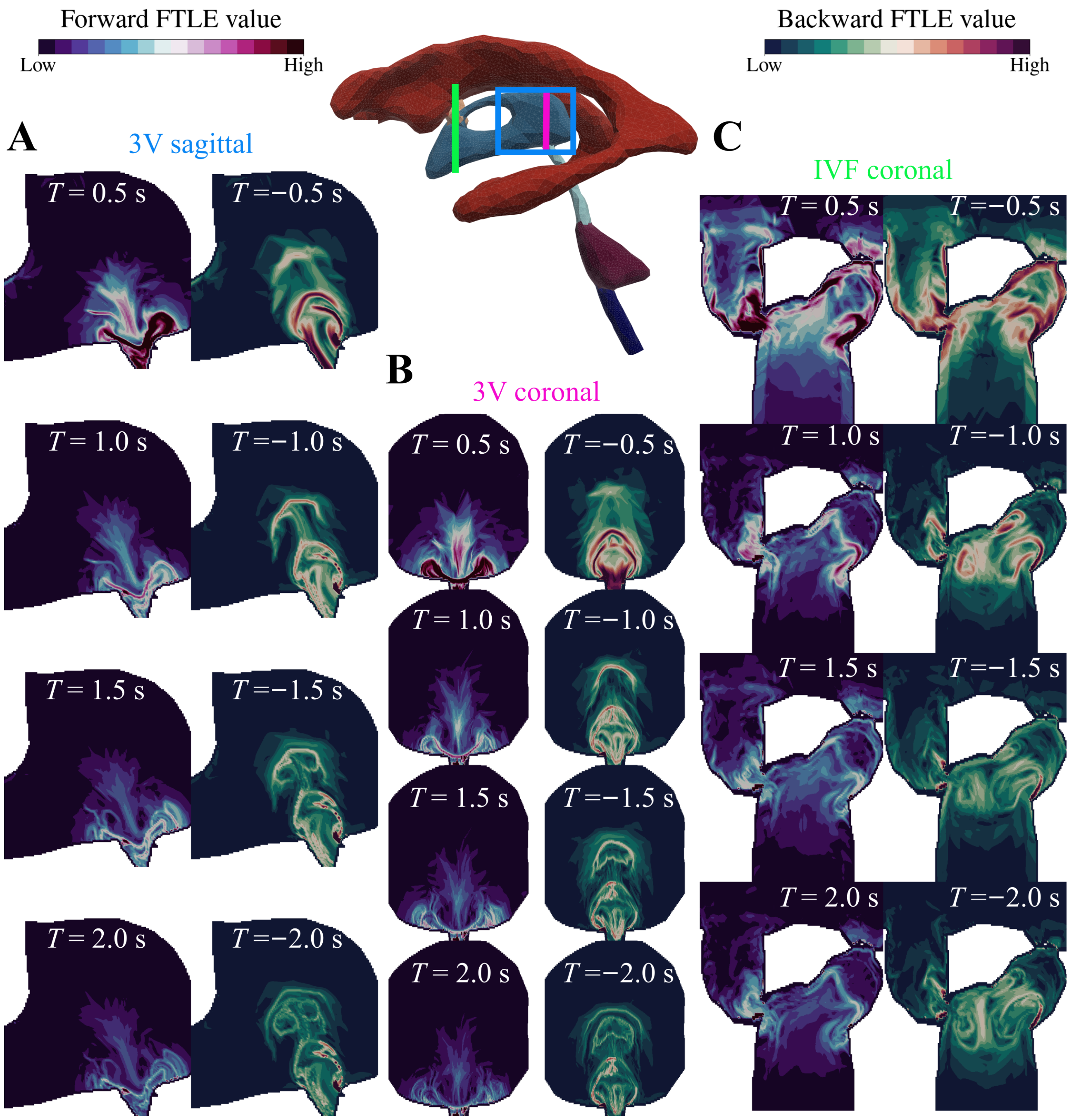

Figure 6: Slices of FTLE fields computed with initial release time $t_0 = 0$ and integration times $T = \pm\{0.5, 1.0, 1.5, 2.0\}$ s in the human brain ventricles. The FTLE fields were computed in two box regions: (i) the posterior end of the third ventricle, including the superior most part of the aqueduct, and (ii) the anterior part of the lateral ventricles, including the interventricular foramina. Sagittal slices in the 3V (**A**) and coronal slices in the 3V (**B**) and the IVF (**C**) of these FTLE fields are displayed. Within each slice-specific column, the left and right columns display forward- and backward-time FTLE fields, respectively. The lower color bar limit is zero in all slices, while the higher color bar limit is an FTLE value of 2.5 and 5.0 for the IVF and 3V slices, respectively.

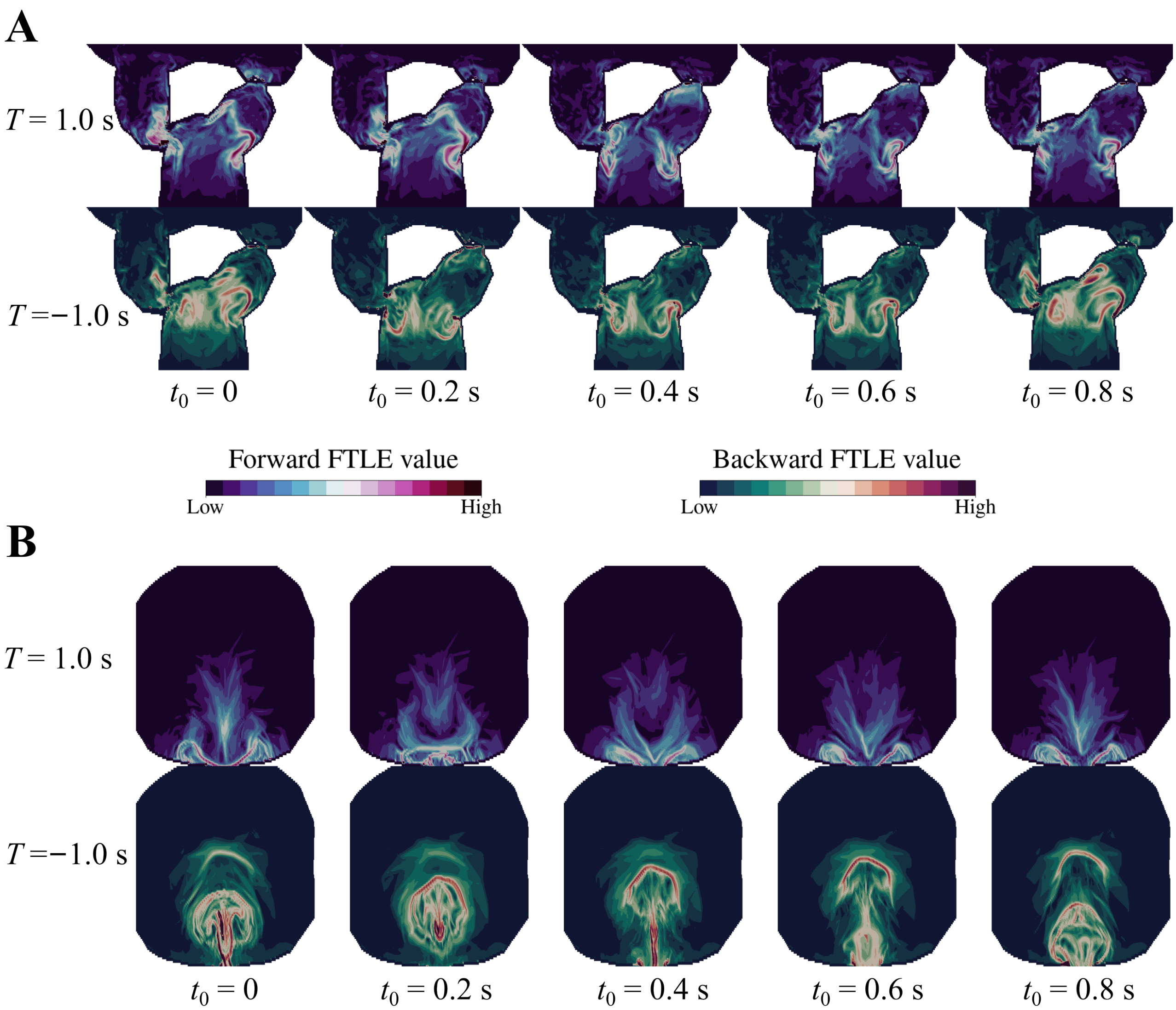

Figure 7: Coronal slices of FTLE fields in the human brain ventricles computed with integration times $T = \pm 1.0$ s (one cardiac cycle). The FTLE fields are computed with varying initial release times t_0. The top two rows are slices taken in the IVF, and of these two rows the top and bottom display forward- and backward-time FTLE fields, respectively. The bottom two rows are sagittal slices taken in the 3V (forward- and backward-time FTLE displayed at the top and bottom).

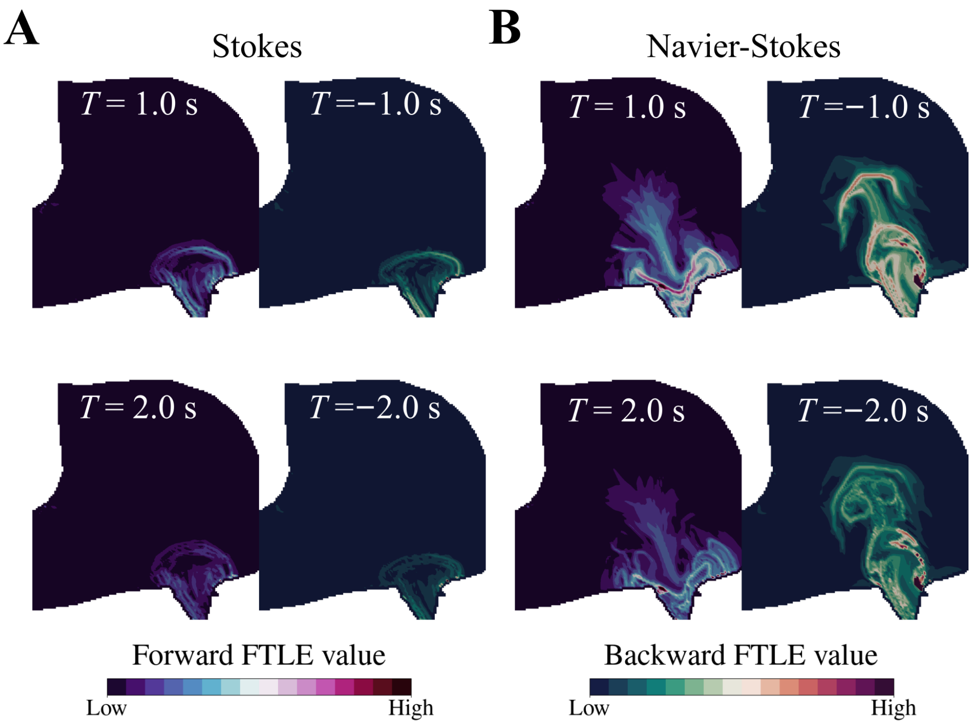

Figure 8: Sagittal slices of FTLE fields computed in the human brain ventricles using the unsteady Stokes equations (**A**) and the Navier-Stokes equations (**B**) with initial particle release time $t_0 = 0$ and integration times $T = \pm\{1.0, 2.0\}$ s. In both subpanels, forward-time FTLE are displayed in the left column while backward-time FTLE are displayed in the right column.

3.7 Brain tissue deformations are the main contributor to establishing Lagrangian structures

The FTLE fields computed in the third ventricle with the DC and DS models are generally comparable to the DSC results, with the formation of a vortex ring (Figure 9). When excluding secretion (DC model), the leading edge of the vortex advances slightly faster (Figure 9A). Meanwhile, excluding cilia (DS model) makes the forward-time FTLE field less attached to the ventricular walls close to the aqueductal entrance (Figure 9B).

3.8 FTLE fields demonstrate brain ventricular flow compartmentalization in zebrafish embryos

Experimental and computational studies have demonstrated compartmentalization of brain ventricular flow in embryonic zebrafish ventricles (Herlyng et al., 2025; Olstad et al., 2019). We therefore investigate structures in the FTLE fields in the interventricular duct connecting the anterior and middle ventricles to see if we can observe such flow compartmentalization in the structures defined by the FTLE fields. As observed in Figure 10, there are clear ridges in the FTLE fields defining attracting and repelling LCS. These repelling and attracting LCS combine to partition regions of distinct flow dynamics. The LCS confine CSF to respective ventricles, demonstrating compartmentalization of flow. The integration time T required to reveal these structures is more than one order of magnitude larger than the T values used for the human ventricles, a result of the lower velocities in the zebrafish brain ventricles.

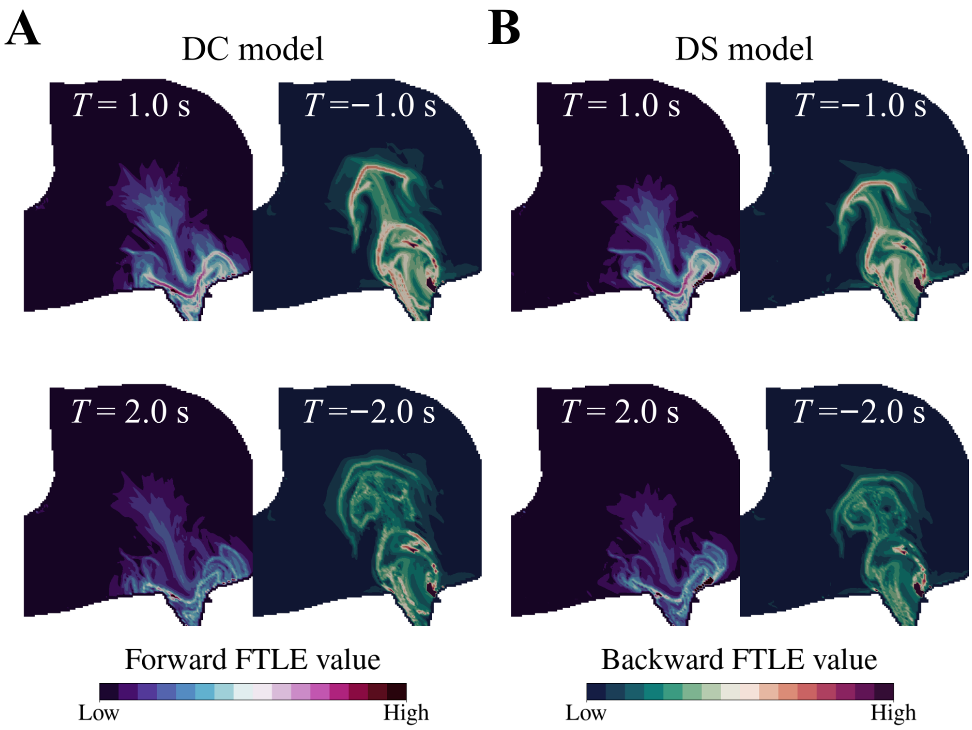

Figure 9: Sagittal slices of FTLE fields computed in the human brain ventricles using the DC model (**A**, no secretion) and DS model (**B**, no cilia) with initial particle release time $t_0 = 0$ and integration times $T = \pm\{1.0, 2.0\}$ s. In both subpanels, forward-time FTLE are displayed in the left column while backward-time FTLE are displayed in the right column.

4 Discussion

4.1 Lagrangian structures are present in the ventricular system

Both in the zebrafish and the human model, we find ridges in the finite-time Lyapunov exponent fields (FTLE) that indicate the presence of Lagrangian coherent structures (LCS), demonstrating organization of the CSF flow. In the human ventricular system, the jet observed in the third ventricle (Figure 6) is consistent with previous computational studies (Kurtcuoglu et al., 2007). The unsteady FTLE fields (Figure 7) indicate that the jet creates a vortex ring every cardiac cycle. As characterized in previous studies, this vortex ring indicates strong stretching and folding of the CSF, contributing to mixing and stirring (Shadden et al., 2006, 2007). The vortex ring also traps fluid, thus the presence of the ring suggests some CSF may occupy the third ventricle for relatively longer times.

For $T = 1.0$ s (one cardiac cycle) we identified a repelling structure in the third ventricle. This repelling structure separates the third ventricle from the aqueduct, which means that once fluid has reached the entrance to the aqueduct, it is quickly expelled through the aqueduct. Meanwhile, fluid inside the third ventricle can stay there for a longer time without exiting through the aqueduct. In the interventricular foramina, there are also clear ridges in the FTLE field with $T = 1.0$ s (Figure 6), indicating regions of high stretching that partition flow with different advective transport times. The lobes that develop have a structure that could trap fluid close to the ventricular walls.

In the zebrafish ventricles FTLE fields (Figure 10), there are clearly defined LCS, both attracting and repelling, in the interventricular duct separating the anterior and middle ventricles. Combining the backward- and forward-in-time FTLE fields, the prominent attracting and repelling structures form a hyperbolic separation point. This confines flow to separate parts of the ventricular system. This finding aligns well with previous studies that have demonstrated flow compartmentalization in embryonic zebrafish brain ventricles (Herlyng et al., 2025; Olstad et al., 2019).

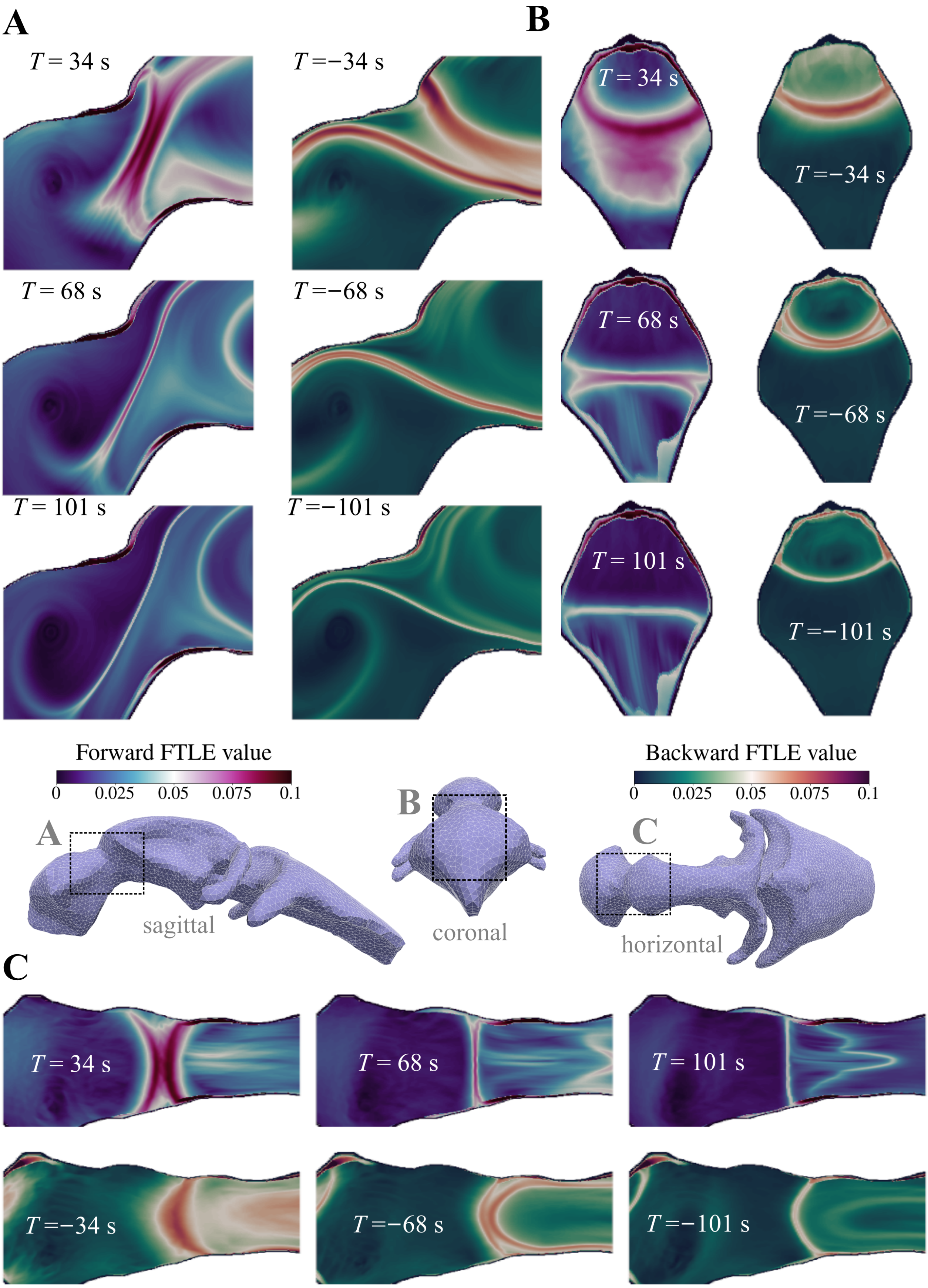

Figure 10: FTLE fields for integration times $T = \pm\{34, 68, 101\}$ s in embryonic zebrafish brain ventricles. The FTLE fields were computed in a box section of the interventricular duct connecting the anterior and middle ventricles. Sagittal (**A**), coronal (**B**) and horizontal slices (**C**) in the center of the box are displayed. Common legends for all forward-in-time and all backward-in-time fields.

4.2 Accounting for fluid inertia is important to resolve Lagrangian structures, but less important for macroscopic flow analysis

The results we present indicate that when simulating CSF flow in human ventricles at Reynolds numbers in the lower hundreds, accounting for inertial effects through solving the full Navier-Stokes equations is important when investigating advective transport by the CSF flow and analyzing local flow features. We draw this conclusion based on the significant differences in the structures present in the FTLE fields (Figure 8) and the differences observed in local flow features (Figure 4) between the Navier-Stokes and Stokes solutions. The lobe structures and vortex rings that form in the Navier-Stokes FTLE fields indicate stronger stirring and mixing of CSF than in the CSF flow approximated with the unsteady Stokes equations.

In terms of computing macroscopic or integrated quantities like stroke volume, however, the unsteady Stokes equations seem sufficiently accurate, showing little discrepancy with the Navier-Stokes solution. As such, the Stokes equations can be utilized as a low-cost alternative for fast predictions in computational frameworks where integrated quantities are needed. We note that several previous computational studies report maximal aqueductal CSF velocities on a mm/s scale (Causemann et al., 2022; Enzmann and Pelc, 1991, 1993; Greitz et al., 1993; Howden et al., 2008; Sweetman and Linninger, 2011; Sweetman et al., 2011; Wagshul et al., 2011), which is ten times lower than the cm/s scale reported here and in several other works (Kim et al., 1999; Kurtcuoglu et al., 2007; Nitz et al., 1992). This ten-fold reduction in maximal velocity would reduce the Reynolds number by a factor ten, yielding for our model Reynolds numbers in the tens instead of the hundreds. Although inertial forces would still dominate viscous forces at these numbers, investigating the effects on the discrepancy in the advective transport between the Stokes and Navier-Stokes solutions would be interesting to address in future work.

4.3 Tissue deformations alter Lagrangian structures more significantly than motile cilia and CSF secretion

Disregarding motile cilia (DS model) did not alter the flow in the human brain ventricles significantly in terms of macroscopic flow quantities like stroke volume. There were however minor differences in the FTLE field structures computed with the DS model compared to both DC and DSC, with the lobe structures near the aqueduct entrance more attached to the 3V wall when including cilia (DC/DSC), indicating that cilia alter the near-wall flow (Figure 9). This finding compares well with the computational study by Siyahhan *et al.* (Siyahhan et al., 2014), whose results showed only a minor effect of cilia on near-wall flow in the lateral ventricles. On the other hand, in the model of Yoshida *et al.* (Yoshida et al., 2022) cilia had a significant effect on mixing. We observed that the cilia model parameters strongly influenced the effect of cilia on the flow (see Section 4.5), and we are of the opinion that any further insights on the effect of cilia in brain ventricular flow and transport would heavily benefit from more experimental data on cilia parameters in the human brain ventricles. This contrasts the case for the embryonic zebrafish brain, where the model parameters are more comprehensively resolved, and previous work strongly suggests cilia are the main drivers of flow (Herlyng et al., 2025; Olstad et al., 2019). The greater effect of cilia in zebrafish ventricles is a direct result of the smaller length scale of the ventricular system of embryonic zebrafish as compared to adult humans. Cilia may play a similarly important role in smaller confined spaces of the human ventricular system, such as in the inferior and occipital horns of the lateral ventricles.

The constant-in-time CSF secretion we imposed had little impact on the computed flow quantities, apart from generating a net outward flow from the ventricular system. In terms of local flow features, these were largely unaffected when disregarding CSF secretion (DC model) compared to the DSC model. In the FTLE fields, we observe a small difference in the location of the leading edge of the vortex ring when disregarding secretion (Figure 9). This may result from the secretion increasing the total CSF volume flowing through the system, leading to a faster development of the vortex ring.

4.4 Model validation with experimental data of brain deformations and ventricular CSF flow

We calibrated the displacement boundary data $\boldsymbol{g}(\boldsymbol{x}, t)$ to ensure a physiological magnitude of CSF flow (Eide et al., 2021; Wagshul et al., 2011). The calibration resulted in a reduction in the magnitude of the displacements, while retaining their qualitative behavior. With our calibrated displacement data, the maximum volume change in our model was around 0.15%. This is less than that reported by previous research, where 0.9% (Zhu et al., 2006) and even as high as 20% (Lee et al., 1989) has been reported. These higher values would produce unrealistically large flow volumes in our model, and the discrepancies between our model and experimental reports indicate the absence of another volume-regulating mechanism in our model. One potential explanation could be that our model lacks the parts of the CSF flow system downstream of the ventricles, and models the median aperture and lateral apertures as open (traction-free) boundaries. In reality, there is some flow resistance since CSF downstream of the ventricular system must be displaced if the ventricular CSF is to be displaced.

The cephalocaudal displacements we imposed are well-documented (Enzmann and Pelc, 1992; Greitz et al., 1992; Kurtcuoglu et al., 2007; Soellinger et al., 2009, 2007), but there is uncertainty regarding displacements in the medial-lateral direction. Our understanding is that arterial swelling exerts pressure on the parenchyma and results in squeezing of the ventricular system, a squeezing which seems to be symmetric in the horizontal plane, as symmetry characterizes medial-lateral motion in some experimental reports (Abderezaei et al., 2021; Soellinger et al., 2009). It has been reported that the periventricular deformations could be related to deformations in the third and lateral ventricles (Abderezaei et al., 2021). We imposed medial-lateral displacements on the third and lateral ventricles based on these reported findings, and the fact that our model was unable to produce physiological CSF flow without these displacements; when imposing only cephalocaudal displacements, the CSF flow rates were less than 10% of experimentally reported values. This resonates well with the conclusion of Kurtcuoglu *et al.* (Kurtcuoglu et al., 2007), namely that medial-lateral displacement of the third ventricle is required to fully explain the observed flow rates of ventricular CSF flow.

We considered the beating direction of cilia to align with the direction of CSF production velocities, similar to Siyahhan *et al.*, who considered directionality of the cilia forcing aligned with the wall shear stress of the macroscopic flow simulated in a production-only setup (Siyahhan et al., 2014). In our setup, with the only free boundaries of the geometry located in the caudal end of the geometry, the production velocity and thus the ciliary beating is generally caudally directed. Assessing the impact of this directionality, and whether e.g. a more cephalic directed beating influences the results, could be investigated in future work.

4.5 Limitations

This modeling work comprises several mechanisms, parameters and assumptions that are uncertain. For the tissue deformation equations, we used a damped linear elastic model where there is an inherent uncertainty in the material parameters of brain tissue (Smillie et al., 2005). Biological soft tissues rarely exhibit Hookean material behavior, and nonlinear models are presumably most appropriate to model deformations over longer time spans (Cheng and Bilston, 2007, 2010; Kaczmarek et al., 1997; Smillie et al., 2005; Tully and Ventikos, 2009). We justify our choice of deformation model by the fact that we seek simply a smooth extension to the rest of the domain boundary for the displacement measurement boundary condition $\boldsymbol{g}(\boldsymbol{x}, t)$, while the ventricles are in reality fluid-filled cavities. The choice of linear elasticity with linear damping is considered adequate since we consider deformations on the timescale of a cardiac cycle, where nonlinear damping effects are negligible.

In terms of the flow model, there are mechanisms of CSF flow that we disregarded. For example, respiration has an influence (Vinje et al., 2019), while we only modeled pulsatile deformations driven by a cardiac frequency. We disregarded the respiration partly because cardiovascular pulsations have been found to be the dominant pulsatile mechanisms in driving CSF flow (Söderström et al., 2025). Other mechanisms like absorption of CSF and neural activity were not modeled because of our focus on the impact of cilia and CSF secretion on the ventricular flow and transport. Additionally, we comment that our model lacks other parts of the CSF flow system such as the spinal canal and the subarachnoid spaces, a choice made based on our focus on ventricular flow.

We model CSF secretion as constant in time and evenly distributed across the choroid plexuses. These assumptions may not perfectly reflect reality, as there are still debates whether CSF may be produced elsewhere in the brain (Miyajima and Arai, 2015). For example, Brodal (Brodal, 2016) report that in reality 10–30% of the CSF production may come from interstitial fluid transported through the ependymal cell layer of the brain ventricles walls into the ventricles. A variable-in-time production of CSF and a different distribution of the production magnitudes between the different choroid plexuses could impact our results.

As already mentioned, there is also uncertainty around our choice of Navier slip boundary condition and the associated parameter values. First, the force per surface area parameter τ is based on the value used by Siyahhan *et al.* (Siyahhan et al., 2014), which they obtained by calibrating CSF flow simulations with rodent CSF velocity data. Because the ependymal cells lining the human brain ventricular walls in general are ciliated to a higher degree than in the rodent brain (Spassky and Meunier, 2017), the ciliary beating might be stronger in human brain ventricles than the beating in the rodent brain. Second, the friction coefficient $\alpha = 47$ Pa $\cdot$ s/m is estimated by assuming a slip length equal to the depth of the cilia layer lining the wall. This was chosen partly by comparing the cilia-generated velocities for different values of α. Setting $\alpha = 0$ like in the zebrafish model, the wall velocities exceeded 1 cm/s, which we considered large compared to measured values and our current understanding of ventricular flow. On the other hand, a far higher value $\alpha = 1000$ Pa $\cdot$ s/m was used by Causemann *et al.* (Causemann et al., 2022) who modeled the ventricular walls as porous with transport into the neighboring parenchyma. However, $\alpha = 1000$ Pa $\cdot$ s/m is approaching a no-slip regime. The resulting particle paths generated would thus likely not compare well with in vivo data (Stadlbauer et al., 2010). We therefore chose the value of $\alpha = 47$ Pa $\cdot$ s/m as a compromise between these two cases. Further testing of values for both τ and α may provide further insights on this.

5 Conclusions and outlook

We have presented a computational framework for using finite-time Lyapunov exponent (FTLE) fields to characterize Lagrangian coherent structures in brain ventricles. The FTLE fields are computed from velocity fields predicted with finite element simulations of cerebrospinal fluid (CSF) flow. Of the three flow mechanisms we considered, cardiovascular pulsation-related deformations, choroid plexus secretion, and motile cilia, the deformations are the main contributor to establishing Lagrangian structures in the brain ventricles. Additionally, we demonstrated that accounting for fluid inertia in flow field predictions has little importance when computing integrated flow quantities, but fluid inertial effects are crucial for capturing Lagrangian coherent structures and thus crucial when modeling transport and mixing in the (human) brain ventricles. Through comparison with Eulerian-based flow analysis of CSF flow in embryonic zebrafish brain ventricles, our results further demonstrate how FTLE fields can be used to characterize prominent flow structures in brain ventricles and partitioning of the flow. An interesting path for future work would be to assess how these structures change with either pathological flow regimes or changes in the ventricular geometry. Although there are still several unresolved model parameters and assumptions, this work hopefully inspires future work on Lagrangian analysis of brain ventricular flow and transport.

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A Model verification

Correctness of the zebrafish CSF flow model implementation was verified in previous work (Herlyng et al., 2025), thus we only present verification analysis of the human brain model here. Mesh and polynomial order refinement studies for the human model were performed both for the fluid and the tissue deformation equations.

A.1 Finite element methods for the fluid equations

Correctness of the discretization schemes on the human brain ventricles geometry were verified using the Method of Manufactured Solutions (Roache, 2001). Convergence properties of the fluid discretization scheme were assessed with two model problems. To assess spatial convergence, we solved a channel flow problem, observing linear convergence in both the velocity error measured in the $\boldsymbol{H}^1$ semi-norm and the pressure error measured in the L^2 norm when employing BDM_1–DG_0 elements. Temporal convergence was assessed with a Taylor vortex problem (Karniadakis and Sherwin, 2005), where the first-order accuracy of the backward Euler scheme was confirmed.

On the human brain ventricles mesh, convergence was assessed for computed quantities of interest like stroke volumes and maximum and minimum velocity magnitudes, by computing these with three different finite element polynomial orders. Comparison of these computations was also carried out on the coarse and the standard human mesh, where consistency in results was observed. We also assessed cardiac cycle independency, and observed a negligible error between simulating for 2 or 3 cardiac cycles, justifying our choice of 2 cycles in the main body of the text.

A.2 Finite element method for the elasticity equations

We refine element polynomial degree on a single mesh. We also refine mesh with a single element degree. We consider three versions of the ventricles mesh, with a total of 51 016 (maximal 0.73 cm, minimal 0.049 cm), 408 128 (maximal 0.40 cm, minimal 0.023 cm), and 3 265 024 (maximal 0.22 cm, minimal 0.010 cm) tetrahedral cells. We look at convergence in the maximum displacement magnitude, the displacement in a point on the traction-free boundary, and the kinetic and elastic energies.

Cardiac cycle independency was assessed. We simulated flow for 2 cardiac cycles and deformation for 4 cardiac cycles. Simulating flow for 4 instead of 2 cardiac cycles showed made virtually no change to most of the computed flow quantities, with the biggest discrepancy being a 1.5% change in the max-to-min pressure gradient, verifying periodicity. The displacement results were also unchanged when simulating for more than 4 cardiac cycles. These results justify our choice.

We verified the implementation of the linear elasticity equations by numerically solving the equations with quadratic basis functions. The errors of the displacement approximation measured in the $\boldsymbol{L}^2$ norm and $\boldsymbol{H}^1$ semi-norm converged at the optimal rates of three and two, respectively. In time, second-order accuracy of the Newmark-β method was confirmed.

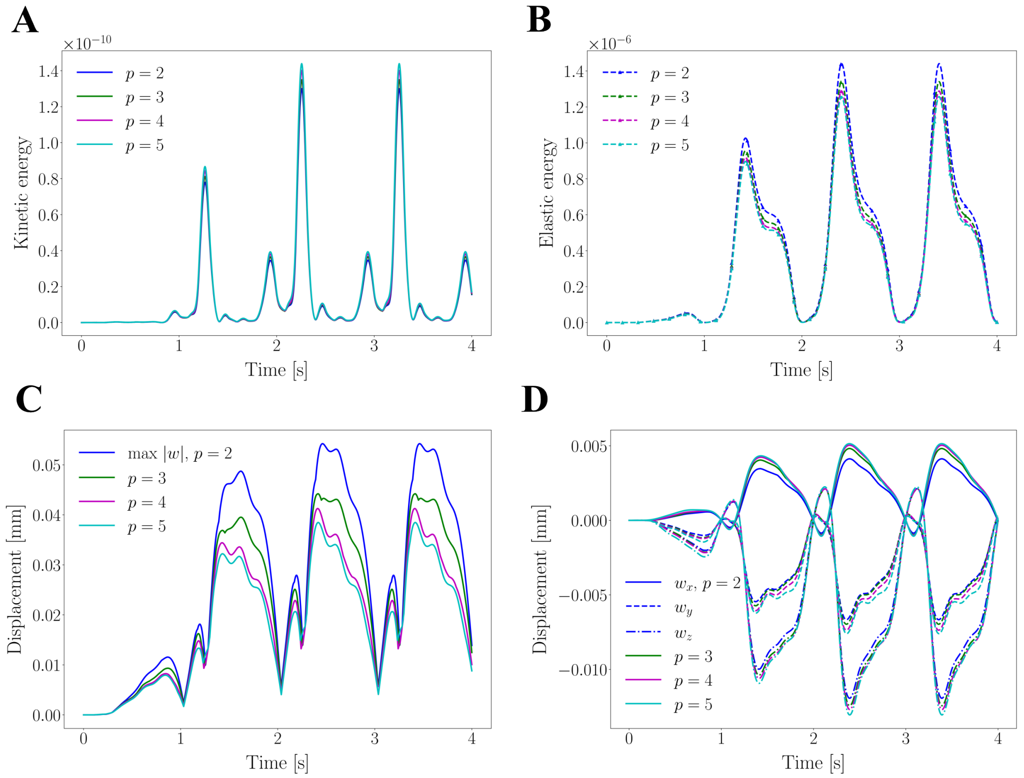

Figure 11: Convergence under element degree (p) refinement for the linear elastodynamics model on the standard mesh (mesh 1).

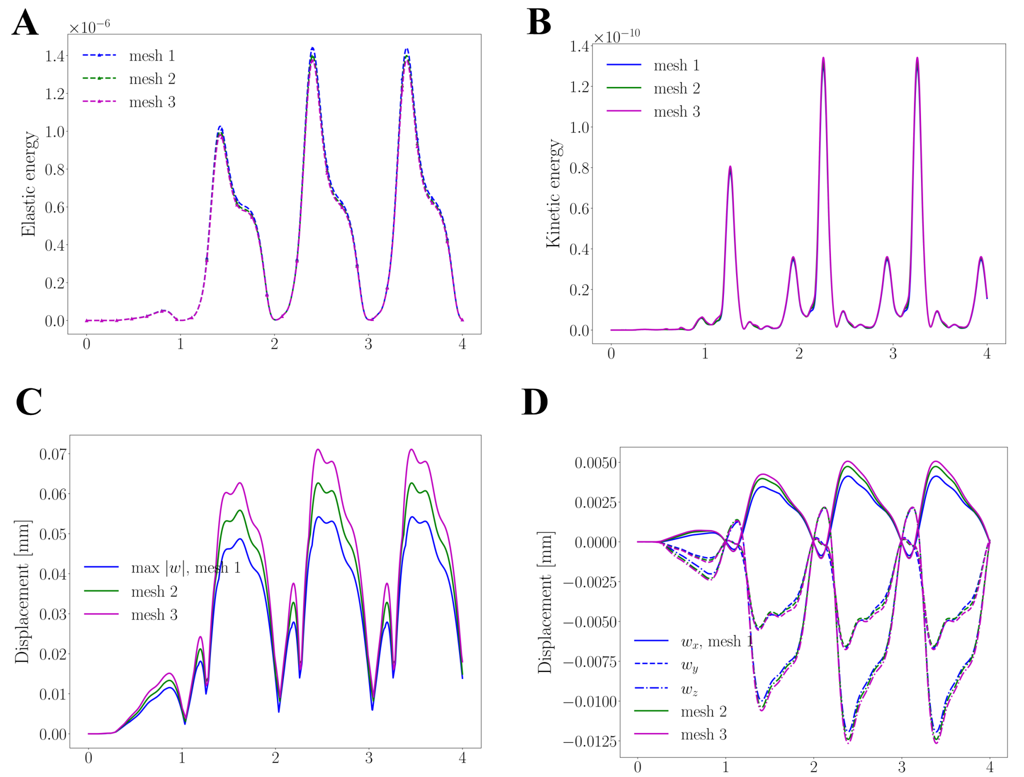

Figure 12: Convergence under mesh refinement for the linear elastodynamics model using quadratic polynomial basis functions ($p = 2$). The standard mesh is mesh 1.